

Proximal Causal Inference: Theory, Methods, and Applications under Unmeasured Confounding

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1 Introduction

Consider influenza vaccination in elderly populations. Observational studies find that vaccinated individuals show a large reduction in the risk of pneumonia or influenza hospitalisation, even after adjusting for measured health indicators. Yet the same analyses find that vaccination is similarly “protective” against hospitalisation for injury or trauma, with a relative risk reduction of roughly 17% [1]. Influenza vaccination, of course, cannot set a broken bone.

The association cannot be causal. It arises from a structural feature of who receives the vaccine: individuals who seek out vaccination tend, on average, to be healthier and less prone to accidental injury, for reasons unrelated to the vaccine itself. This produces a systematic difference between vaccinated and unvaccinated populations that persists across every outcome the study measures. Because health-seeking behaviour drives both vaccination uptake and a wide range of health outcomes, standard covariate adjustment cannot remove this bias when the relevant behaviour is not fully measured [2]. As a result, such variables (latent health status, for instance) are routinely absent from administrative records and clinical surveys: the confounders that matter most in observational data are often precisely those that are hardest to record. Observational evidence of this kind informs drug approvals, clinical guidelines, and public health resource allocation; we read the systematic nature of the failure as making this a consequential problem with implications beyond methodology.

A long-standing epidemiological response, formalised by Lipsitch et al. [2] with earlier antecedents in Rosenbaum [3, 4], is to exploit *negative control* variables. A *negative control outcome* is an outcome known not to be caused by the treatment of interest; a *negative control exposure* is a variable known not to causally affect the outcome. Both types are selected to share the confounding mechanism of the primary treatment–outcome relationship. Any observed association between a negative control outcome and the treatment, or between a negative control exposure and the outcome, is evidence that unmeasured confounding is present. In the vaccine study, injury hospitalisation serves this role: its association with vaccination, precisely because it cannot be causal, signals the confounding. The 17% reduction is informative beyond its diagnostic value; it measures the magnitude of the selection effect, indicating how strongly health-seeking behaviour separates vaccinated from unvaccinated individuals. Beyond signalling confounding, the negative control outcome carries quantitative information about the unmeasured confounder.

Indeed, the diagnostic use of auxiliary variables leads to identification rather than mere diagnosis. A negative control outcome is correlated with the primary outcome precisely because both are shaped by the same latent confounder U ; a negative control exposure is correlated with the treatment for the same structural reason. It follows that the two auxiliary variables carry information about U . Miao et al. [5] established that, when a treatment-side proxy and an outcome-side proxy each satisfy an independence condition and a rank (completeness) requirement, the average causal effect can be recovered from observed data alone, without modelling how the proxies relate to the latent confounder and without specifying the measurement error process. Identification proceeds through a pair of integral equations. These define the *bridge functions* used throughout the proximal framework.

Shi et al. [6] established multiply robust inference for the categorical-confounder case. Building on this result, Tchetgen Tchetgen et al. [7] formalised *proximal causal inference* (PCI) as a unifying programme and introduced the vocabulary of *treatment confounding proxies* and *outcome confounding proxies*. The semiparametric theory for the general nonparametric case was established by Cui et al. [8], who derived the efficient influence function for the average causal effect, proved double robustness, and characterised the rate conditions under which semiparametric efficiency is achievable; these results form the theoretical foundation of this essay. In parallel, Kallus et al. [9] proposed minimax bridge estimators. A complementary contribution from Bennett et al. [10] then clarified that root- n inference on the causal effect requires a *source condition* on the target functional relative to the inverse operator, not a fast convergence rate for the bridge itself. Most recently, Bennett et al. [11] examined spectral source conditions for bridge inverse problems and the L^2 convergence guarantees they yield, even without the closedness assumption that earlier work required.

Proximal causal inference provides a route from observational data to causal conclusions without modelling the unmeasured confounder, provided two proxy variables satisfy certain regularity conditions and the bridge functions can be estimated reliably. The question we pursue is whether this theoretical apparatus translates into estimators that are accurate, well-calibrated, and robustly tunable. We develop the framework from identification through to inference, implement it in four estimators of increasing generality, and subject these to Monte Carlo stress-tests designed to identify the regimes in which they remain reliable; the conclusions address method selection, proxy requirements, and the interpretation of causal estimates in observational studies.

The remainder of the essay is organised as follows. Section 2 formalises the observational setting and the six identification conditions under which the causal effect is recoverable through proxies. The theoretical core then follows: Section 3 introduces the bridge functions and the four-layer estimation hierarchy; Section 4 derives the efficient influence function and establishes double robustness; Section 5 quantifies the spectral regularity that governs inference under regularisation and applies the framework to the noisy-binary-treatment case. Building on this theory, Section 6 derives the finite system solved by each of the four bridge methods and introduces the calibration metrics; the blind-tuning protocol that uses them is developed in Section 7. Section 7 reports Monte Carlo experiments across a binary and two continuous data-generating processes, together with a proxy quality grid. These simulations validate the spectral predictions of Section 5, build and stress-test the protocol on the diagnostics of Section 6, and produce a regime map for method selection. Section 8 then applies this protocol to two real-data benchmarks (return to schooling in the NLSY79 cohort, $N = 4977$; effect of right-heart catheterisation on 30-day mortality, $N = 5735$), triangulating proximal estimates against external literature anchors.

2 The Proximal Setting

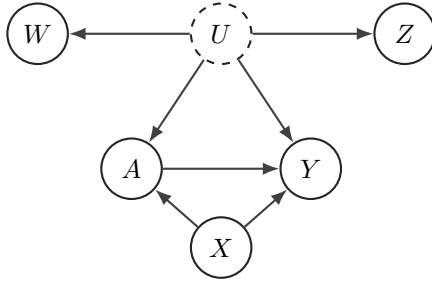
2.1 The proximal causal model

This section formalises the proximal causal model, following Miao et al. [5], Tchetgen Tchetgen et al. [7], and Cui et al. [8]. We then lay out the six identification assumptions that license proximal recovery and introduce the estimand that serves as the working target throughout. The model augments the standard observational study with two auxiliary variables that share the unmeasured confounder and are restricted by structural independence conditions of their own; Figure 1 renders the structure and makes visible the central identification difficulty.¹

The treatment A and outcome Y are connected by two paths: the direct edge $A \rightarrow Y$, which carries the causal effect of interest, and the back-door route $A \leftarrow U \rightarrow Y$, along which the unmeasured confounder U generates an associational dependence with no causal content. Standard regression of Y on A combines both contributions, producing precisely the bias displayed in the vaccination study of Section 1. The proximal model provides two further observed variables, the proxies W and Z , each connected to the outcome and treatment sides only through the latent U . The proxies make the back-door path readable: comparing how W and Z relate to A and to Y recovers the structure of the unmeasured confounding without ever observing U .

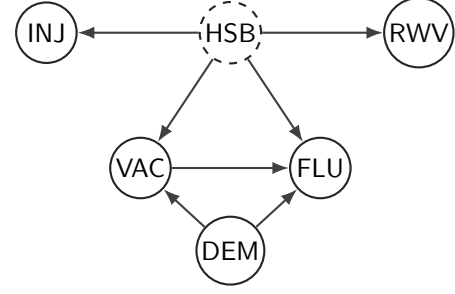
In the influenza application of Figure 1b, the unobserved confounder is health-seeking behaviour, a latent disposition that shapes both the propensity to vaccinate and a wide range of health outcomes. A natural outcome-side proxy is the rate of injury hospitalisation, which cannot be caused by the vaccine but is shaped by the same latent disposition [2]. On the treatment side, a behavioural indicator with no biological pathway to influenza outcomes plays an analogous role: ringworm-related healthcare visits have been used in this position [12].

¹We observe n i.i.d. copies of $O = (Y, A, W, Z, X)$, with $A \in \mathcal{A} \subseteq \mathbb{R}$ continuous (the binary case is a special instance). Measured covariates X are suppressed when they play no role in the argument; every conditional statement is then understood to hold conditional on X . All expectations are taken under the joint law $\mathbf{P}$ of O .



(a) Formal model.

U : latent confounder; W : outcome proxy; Z : treatment proxy;
 A : treatment; Y : outcome; X : observed covariates.



(b) Vaccination instance.

HSB: health-seeking behaviour; INJ: injury hospitalisation;
RWV: ringworm-related visits; VAC: vaccination;
FLU: influenza hospitalisation; DEM: age, sex, comorbidities.

Figure 1: The proximal causal model in the formalism of this essay (a) and its instantiation in the influenza vaccination example (b). The unobserved variable is drawn with a dashed circle. Each absent edge encodes a structural exclusion restriction on W and Z .

2.2 Identification assumptions

The assumptions that license proximal identification organise into a familiar sequence. We list six: a link between potential and observed outcomes, a positivity condition, a latent unconfoundedness condition, two exclusion restrictions on the proxies, and a rank condition that ensures the proxies carry enough information about the latent confounder.

Assumption 2.1 (Consistency). $Y = Y(A)$ almost surely.

Assumption 2.2 (Positivity; Cui et al. [8], Kallus et al. [9]). $0 < f(A | U, X) < \infty$ for almost all (A, U, X) .

Assumption 2.3 (Latent Unconfoundedness; Tchetgen Tchetgen et al. [7]). $Y(a) \perp\!\!\!\perp A | U, X$ for all $a \in \mathcal{A}$.

Assumption 2.4 (Outcome Proxy Independence; Miao et al. [5]). $W \perp\!\!\!\perp (A, Z) | U, X$.

Assumption 2.5 (Treatment Proxy Independence; Miao et al. [5]). $Z \perp\!\!\!\perp Y | U, A, X$.

Assumption 2.6 (Completeness of W ; Miao et al. [5], Cui et al. [8]). For any $g \in L^2$, $\mathbb{E}[g(U) | W, A, X] = 0$ a.s. implies $g(U) = 0$ a.s.

We discuss each in turn. Assumption 2.1 bridges potential and observed outcomes: the value of Y recorded for an individual must coincide with the potential outcome under the treatment they actually received. The formal implications of this assumption for a continuously recorded treatment are examined in Section 5.

Assumption 2.2 requires that every treatment level of interest be reachable within every stratum of the latent confounder; without this, no procedure can estimate the effect at that level without extrapolating beyond the support of the data.

Assumption 2.3 places the unmeasured mechanism between treatment and potential outcomes entirely on U . The condition is the unconfoundedness assumption of observational studies, applied to the latent (U, X) rather than to observed covariates alone. In the vaccination example, it asserts that once health-seeking behaviour and demographic covariates are fixed, vaccination uptake carries no remaining information about the potential outcomes.

Assumptions 2.4 and 2.5 are the exclusion conditions on the proxy variables: Assumption 2.4 closes any direct path from W to A or Z (any empirical association between W and the treatment side runs through U alone), while Assumption 2.5 closes any direct effect of Z on Y (ringworm-related visits correlate with influenza hospitalisation only via their shared dependence on U). These are the structural restrictions a practitioner accepts in proposing a proxy pair.

Assumption 2.6 requires that the proxies carry enough information about U to recover the confounding structure: no nontrivial function of U may be invisible to W given (A, X) . In the vaccination example, the condition asks that distinct profiles of health-seeking behaviour produce distinguishable distributions over injury hospitalisation rates; were two latent profiles to induce identical injury patterns, the proxy could not separate them and the confounding mechanism would remain hidden.

Under assumptions 2.1 to 2.6, the absent edges in Figure 1 express the design discipline of the proximal study: each missing arrow is a conditional independence to be defended on subject-matter grounds. Identifying the causal effect from these conditions is one matter; estimating it from finite samples is harder, and the focus of Section 3.

2.3 The estimand

For a binary treatment, the natural causal estimand is the average treatment effect $\mathbb{E}[Y(1) - Y(0)]$. For a continuous treatment, the corresponding object is the dose-response curve

$$m(a) = \mathbb{E}[Y(a)], \quad a \in \mathcal{A}, \quad (2.1)$$

of which the binary effect is a contrast at two points.

Kallus et al. [9] embed both in the broader framework that we adopt throughout. A treatment policy is a probability distribution $\pi(\cdot | X)$ over $\mathcal{A}$ given observed covariates, representing a hypothetical randomised assignment rule. The corresponding causal estimand is the expected outcome under that policy,

$$J(\pi) = \mathbb{E}_\pi[Y] = \mathbb{E}\left[\int Y(t) \pi(dt | X)\right], \quad (2.2)$$

the *generalised average causal effect* (GACE). In particular, the standard estimands are instances: for binary $\mathcal{A} = \{0, 1\}$, the point-mass policies δ_1 and δ_0 recover the average treatment effect $J(\delta_1) - J(\delta_0) = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)]$. The GACE thus subsumes classical estimands within a single causal target parameterised by the choice of policy.

Assumption 2.1 requires that observing $A = a$ be equivalent to intervening to set $A = a$, a requirement that is rarely safe to assume for a continuously recorded treatment. VanderWeele and Hernán [13] show that a single recorded value $A = a$ typically covers several physically distinct interventions, only one of which is realised for any given individual: differences in dose, in delivery, in timing, or in individual absorption all collapse onto the same recorded value. In the vaccination example, “ $A = 1$ ” aggregates different vaccine strains, injection sites, times of year, and individual immune responses; “ $A = 0$ ” aggregates a similarly heterogeneous set of non-treatment trajectories. A point-mass policy at a then targets an idealised intervention of perfectly precise dose a . In response, stochastic-intervention frameworks [14, 15] replace point-mass intervention with a distribution of plausible delivery realisations.

We accordingly take as our working estimand the GACE under a kernel policy at fixed bandwidth. For a target dose a and a fixed bandwidth $b > 0$, the kernel policy $\pi_{a,b}$ has density

$$\pi_{a,b}(t | X) = K_h(t - a), \quad K_h(\cdot) = b^{-1} K(\cdot/b), \quad (2.3)$$

with K a symmetric probability density. Substituting into Equation (2.2) gives

$$J(\pi_{a,b}) = \mathbb{E}\left[\int Y(t) K_h(t - a) dt\right]. \quad (2.4)$$

The bandwidth b is held fixed throughout, at a value chosen by the investigator on substantive grounds: it represents the unavoidable uncertainty with which a prescribed dose is actually delivered, and the recorded value a should accordingly be read as an average over realisations within a neighbourhood of width b . This is a deliberate departure from the standard kernel-smoothing practice in which b is a vanishing nuisance parameter selected from data; here, the bandwidth is part of the estimand definition. The choice has a

further benefit beyond its substantive realism: working with a smoothed target rather than the pointwise dose-response regularises the ill-posed inverse problem at the heart of estimation, an advantage made precise in Section 5.

For a symmetric kernel K with $\mu_2(K) = \int t^2 K(t) dt$, a Taylor expansion at a gives

$$J(\pi_{a,b}) = m(a) + \frac{b^2}{2} \mu_2(K) m''(a) + o(b^2), \quad (2.5)$$

so $J(\pi_{a,b}) \rightarrow m(a)$ as $b \rightarrow 0$, with the gap at fixed $b > 0$ governed by the local curvature $m''(a)$.

$J(\pi_{a,b})$ is the working target throughout; Section 3 shows that, under assumptions 2.1 to 2.6, it is characterised by a pair of bridge equations whose solutions identify the causal effect and from which estimation proceeds.

3 Bridges and Estimation

Under the identification assumptions of Section 2, the working estimand $J(\pi_{a,b})$ is characterised by a pair of bridge equations. The diagnostic use of negative controls is by now well established [2, 5]; the bridge equations promote this diagnostic into an estimation problem. We organise the estimation hierarchy in four layers: parametric two-stage least squares (Section 3.2), kernel mean embedding and KPV (Section 3.3.1), minimax formulation and PMMR (Section 3.3.2), and the doubly robust functional of Section 4.

3.1 The bridge equation and completeness

Define the conditional expectation operator $T : L^2(W, A, X) \rightarrow L^2(Z, A, X)$ by $(Th_0)(Z, A, X) = \mathbb{E}[h_0(W, A, X) \mid Z, A, X]$. The outcome bridge h_0 is any element of $L^2(W, A, X)$ solving

$$Th_0 = \mathbb{E}[Y \mid Z, A, X], \quad (3.1)$$

introduced by Miao et al. [5]. The equation is a Fredholm integral equation of the first kind in h_0 : an observable conditional mean appears on the right, and recovering h_0 requires inverting T , a generically ill-posed problem.

Recall the dose-response identity $\mathbb{E}[Y(t)] = \mathbb{E}[h_0(W, t, X)]$, valid under Assumptions 2.3 to 2.5 [5, 7], where the expectation is over the joint marginal of (W, X) . Combined with the kernel-policy definition (2.4) of Section 2.3, this yields

$$J(\pi_{a,b}) = \mathbb{E} \left[\int h_0(W, t, X) K_h(t - a) dt \right]. \quad (3.2)$$

The difficulty of identification is concentrated in solving (3.1), after which (3.2) is a straightforward integration; Figure 2 contrasts this solve-then-marginalise route with naive regression.

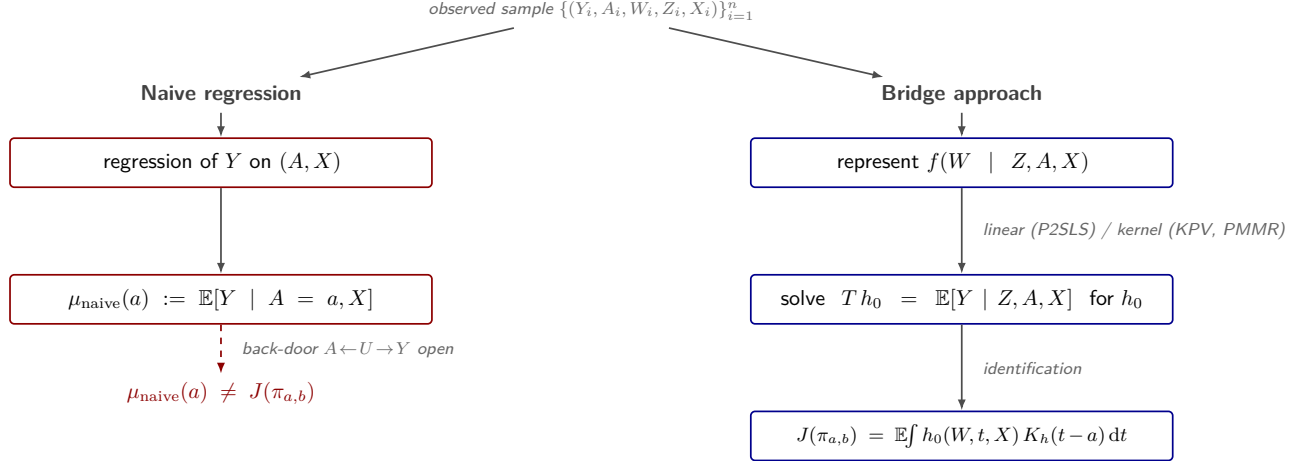


Figure 2: Two strategies for recovering $J(\pi_{a,b})$ from observational data. Direct regression of Y on (A, X) returns $\mu_{\text{naive}}(a)$, which under unmeasured confounding does not coincide with the causal target $J(\pi_{a,b})$ because the back-door path $A \leftarrow U \rightarrow Y$ remains open. The proximal bridge approach exploits the proxies W and Z in three steps: represent the conditional distribution of W , solve the bridge equation, and marginalise. Section 3.2 develops the linear (P2SLS) implementation; Section 3.3 develops the kernel (KPV, PMMR) implementation.

Without further structure, two distinct functions can satisfy (3.1), in which case (3.2) fails to define $J(\pi_{a,b})$ unambiguously. A solution exists provided $\mathbb{E}[Y | Z, A, X]$ lies in $\text{Range}(T)$, which holds under mild conditions on the proxy distribution [16]; uniqueness then requires $\text{Ker}(T) = \{0\}$, which is precisely Assumption 2.6. Instrumental-variable identification relies on an analogous rank condition, there verifiable via the first-stage F -statistic [17]; the proximal completeness condition admits no such diagnostic, and the obstruction is fundamental.

Canay et al. [18] prove that no uniformly consistent test of completeness exists, and the obstruction is a continuity phenomenon. For any joint law $\mathbb{P}_0$ under which W is bounded-complete given (Z, A, X) , one can construct a law $\mathbb{P}_1$ arbitrarily close to $\mathbb{P}_0$ in total variation distance such that completeness fails under $\mathbb{P}_1$. In other words, two regimes that differ qualitatively on completeness are indistinguishable on a finite sample. More precisely, any test of level α that rejects $\mathbb{P}_1$ therefore rejects $\mathbb{P}_0$ with probability at most $\alpha + \epsilon$: its power against completeness violations cannot exceed its size. The condition must therefore be invoked on subject-matter grounds. D’Haultfœuille [16] shows it holds whenever the conditional density of W given (Z, A, X) admits a polynomial expansion in a continuous regressor, which covers Gaussian and exponential-family proxies.

3.2 Parametric estimation

We begin with the simplest case. Restricting h_0 to a parametric class converts (3.1) into a finite-dimensional estimation problem; this is the linear branch of Figure 2. Postulate

$$h_0(W, A, X) = \beta_0 + \beta_W^\top W + \beta_A A + \beta_X^\top X \quad (3.3)$$

for an unknown parameter $\beta = (\beta_0, \beta_W, \beta_A, \beta_X)$. The bridge equation (3.1) becomes a linear moment condition in β , which by linearity of the conditional expectation rearranges into

$$\mathbb{E}[Y | Z, A, X] = \beta_0 + \beta_W^\top \mathbb{E}[W | Z, A, X] + \beta_A A + \beta_X^\top X. \quad (3.4)$$

Equation (3.4) suggests a two-stage procedure: estimate $\mathbb{E}[W | Z, A, X]$, then regress Y on the estimate.

Stage 1. Linear regression of W on (Z, A, X) . Estimate

$$W = \theta_0 + \theta_Z^\top Z + \theta_A A + \theta_X^\top X + \epsilon_W, \quad (3.5)$$

producing fitted values $\hat{W} = \hat{\theta}_0 + \hat{\theta}_Z^\top Z + \hat{\theta}_A A + \hat{\theta}_X^\top X$ that estimate $\mathbb{E}[W | Z, A, X]$.

Stage 2. *Linear regression of Y on $(\hat{W}, A, X)$.* Estimate

$$Y = \beta_0 + \beta_W^\top \hat{W} + \beta_A A + \beta_X^\top X + \epsilon_Y, \quad (3.6)$$

yielding $\hat{\beta}$.

For a linear bridge integrated against a symmetric kernel with $\int t K_h(t - a) dt = a$, substituting $\hat{\beta}$ into (3.2) yields a closed form in the second-stage coefficients:

$$\hat{J}(\pi_{a,b}) = \hat{\beta}_0 + \hat{\beta}_W^\top \hat{\mathbb{E}}[W] + \hat{\beta}_A a + \hat{\beta}_X^\top \hat{\mathbb{E}}[X], \quad (3.7)$$

with empirical analogues for the marginal expectations. Tchetgen Tchetgen et al. [7] term this procedure proximal two-stage least squares (P2SLS).

The completeness condition appears concretely as an identifiability requirement on the moment matrix of the bridge basis: the two-stage procedure returns a unique $\hat{\beta}$ only when this matrix is invertible [5, 8]. In linear and near-linear specifications, this rank condition is directly diagnosable from the design [7].

P2SLS extends to generalised linear models with link-appropriate adaptations in the second stage [19],² at the cost of possible bias under misspecification, a limitation that motivates Section 3.3.

3.3 Nonparametric estimation

Nonparametric estimation of h_0 (the kernel branch of Figure 2) imposes no functional form on the bridge. A kernel ridge regression of Y on (W, A, X) targets the conditional mean of Y given observed inputs, whereas (3.1) requires integrating $h_0(W, A, X)$ against the full conditional distribution $f(W | Z, A, X)$.

3.3.1 Kernel mean embedding and KPV

Definition 3.1 (Kernel mean embedding and conditional mean embedding). *Let k_W be a kernel on W with feature map $\phi_W : W \rightarrow \mathcal{H}_W$ and reproducing kernel Hilbert space $\mathcal{H}_W$. For any distribution ρ on W , the kernel mean embedding is*

$$\mu_\rho = \int \phi_W(W) d\rho(W) \in \mathcal{H}_W; \quad (3.8)$$

for the conditional distribution of W given (Z, A, X) , the conditional mean embedding is $\mu_{W|Z,A,X} = \mathbb{E}[\phi_W(W) | Z, A, X]$ [20].

Under the assumption that k_W is *characteristic* [20], the map $\rho \mapsto \mu_\rho$ is injective: a distribution is uniquely determined by its embedding.

Proposition 3.1 (Representer identity and embedded bridge equation). *For any $h \in \mathcal{H}_W$,*

$$\mathbb{E}[h(W) | Z, A, X] = \langle h, \mu_{W|Z,A,X} \rangle_{\mathcal{H}_W}. \quad (3.9)$$

Applied to the bridge equation (3.1), with (A, X) absorbed into a product kernel so that h_0 lives in a joint RKHS $\mathcal{H}$, this identity rewrites the bridge equation as

$$\langle h_0, \mu_{W|Z,A,X} \rangle_{\mathcal{H}} = \mathbb{E}[Y | Z, A, X]. \quad (3.10)$$

This is the central insight of KPV: the original integral equation reduces to a regression in the space of conditional mean embeddings. Mastouri et al. [20] decompose the resolution of (3.1) into two kernel ridge regressions.

²Liu et al. [19] cover identity link (continuous), log link (count), and logit link (binary); Tchetgen Tchetgen et al. [7] had earlier noted the binary case (Remark 7, p. 384).

Stage 1. Estimate the conditional mean embedding $\mu_{W|Z,A,X}$. Using a sample $\{(W_i, Z_i, A_i, X_i)\}_{i=1}^{n_1}$, solve

$$\hat{\mu}_{W|Z,A,X} \in \arg \min_{\mu} \frac{1}{n_1} \sum_{i=1}^{n_1} \|\phi_W(W_i) - \mu(Z_i, A_i, X_i)\|_{\mathcal{H}_W}^2 + \lambda_1 \|\mu\|_{\mathcal{H}}^2. \quad (3.11)$$

Stage 2. Estimate h_0 in the embedded space. Using a sample $\{(Y_j, Z_j, A_j, X_j)\}_{j=1}^{n_2}$, solve

$$\hat{h} \in \arg \min_{h \in \mathcal{H}} \frac{1}{n_2} \sum_{j=1}^{n_2} (Y_j - \langle h, \hat{\mu}_{W|Z_j, A_j, X_j} \rangle_{\mathcal{H}})^2 + \lambda_2 \|h\|_{\mathcal{H}}^2. \quad (3.12)$$

The two-stage procedure estimates h_0 nonparametrically but has two limitations relevant for downstream causal inference. First, the moment condition is enforced only along directions spanned by the second-stage RKHS; violations of the bridge equation (3.1) along orthogonal directions remain invisible to the criterion [20, 9], which motivates the minimax formulation introduced below. Furthermore, the second stage uses the first-stage estimate $\hat{\mu}$ as input, and the resulting plug-in bias does not in general vanish at the parametric rate when the bridge is plugged into a downstream causal functional; this point is taken up in Section 3.4 and addressed by the doubly robust theory of Section 4.

3.3.2 Minimax formulation and PMMR

The moment condition can be enforced against all directions in $\mathcal{Q}$ simultaneously by reformulating the bridge equation as a minimax problem. For any class $\mathcal{Q}$ of test functions on (Z, A, X) that is dense in L^2 , the pointwise moment condition

$$\mathbb{E}[h_0(W, A, X) - Y | Z, A, X] = 0 \quad (3.13)$$

is equivalent to

$$\mathbb{E}[q(Z, A, X)(h_0(W, A, X) - Y)] = 0 \quad \text{for every } q \in \mathcal{Q}. \quad (3.14)$$

The equivalence follows from density of $\mathcal{Q}$ in L^2 . Condition (3.14) is the variational reformulation of (3.1): the pointwise constraint on (Z, A, X) is replaced by an integrated moment condition against every $q \in \mathcal{Q}$.

The bridge estimate is the minimiser of the worst-case residual over $\mathcal{Q}$:

$$h_0 \in \arg \min_{h \in \mathcal{H}} \sup_{q \in \mathcal{Q}} \mathbb{E}[q(Z, A, X)(h(W, A, X) - Y)]. \quad (3.15)$$

The optimising q identifies the direction of maximal moment violation; the bridge estimate is the function for which no test in $\mathcal{Q}$ produces a large residual. Unlike KPV, which enforces the residual condition along the span of the second-stage RKHS, the minimax formulation tests against every $q \in \mathcal{Q}$.

The empirical analogue of (3.15) is not directly usable. The bilinear functional $\mathbb{E}_n[q(h_0 - Y)]$ is linear in q , so its supremum over an unconstrained $\mathcal{Q}$ is unbounded. Kallus et al. [9] resolve this by introducing a quadratic stabiliser $-\lambda_q \mathbb{E}_n[q(Z, A, X)^2]$ that renders the supremum finite: the inner maximisation over q becomes a strictly concave quadratic with a well-defined closed-form solution. A complexity penalty $\lambda_h \|h\|_{\mathcal{H}}^2$ on the bridge controls estimator complexity. The proximal maximum moment restriction (PMMR) estimator solves

$$\hat{h} \in \arg \min_{h \in \mathcal{H}} \sup_{q \in \mathcal{Q}} \left\{ \mathbb{E}_n[q(Z, A, X)(h(W, A, X) - Y)] - \lambda_q \mathbb{E}_n[q(Z, A, X)^2] \right\} + \lambda_h \|h\|_{\mathcal{H}}^2. \quad (3.16)$$

In (3.16), the bilinear term $\mathbb{E}_n[q(h_0 - Y)]$ vanishes when h_0 satisfies the bridge equation against every $q \in \mathcal{Q}$. The two penalty terms then serve distinct purposes: the stabiliser $-\lambda_q \mathbb{E}_n[q^2]$ does not introduce bias in the population objective and is held fixed as n grows, while the RKHS regulariser $\lambda_h \|h_0\|_{\mathcal{H}}^2$ introduces finite-sample bias but vanishes with n to ensure consistency [9].

Importantly, the function q acts as a test function dualising the moment restriction; a treatment-side bridge enters only with the doubly robust theory of Section 4.

3.4 From plug-in to doubly robust

KPV and PMMR yield $\hat{h}$ at an L^2 rate governed by the ill-posedness of T , which is non-parametric and generally slower than $\sqrt{n}$ [8, 10]. A plug-in estimator $J(\pi_{a,b}; \hat{h})$ inherits this rate, and $\sqrt{n}$ -consistency of the causal functional is not attainable without a correction. We adopt a Neyman-orthogonal estimator based on the efficient influence function derived for the proximal setting by Cui et al. [8]; Section 4 develops the doubly robust theory together with the Kallus–Bennett minimax extensions, and Section 6 details the specific estimators.

4 Doubly Robust Semiparametric Inference

The plug-in estimator $\hat{J}_{\text{REG}} = \mathbb{P}_n[T_\pi \hat{h}]$ of Section 3.4 inherits the nonparametric rate of $\hat{h}$, which under ill-posed bridge equations rules out $\sqrt{n}$ -asymptotic normality and the semiparametric efficiency that the structural assumptions of Section 2.2 would otherwise support. General semiparametric theory provides a canonical answer to both problems: the efficient influence function.

4.1 The plug-in and the EIF construction

The plug-in’s bias is a linear functional of the bridge error:

$$\mathbb{E}[T_\pi \hat{h}] - J(\pi_{a,b}) = \langle \alpha_\pi, \hat{h} - h_0 \rangle_{L^2(W,A,X)}, \quad (4.1)$$

where α_π is the unique element of $L^2(W, A, X)$ that represents the linear form $h \mapsto \mathbb{E}[T_\pi h]$. A small error in $\hat{h}$ propagates linearly to the functional; in an ill-posed problem this first-order sensitivity is what we mainly want to control. Semiparametric theory provides the canonical score, which we now construct.

A familiar instance is the augmented inverse-propensity-weighted estimator for the average treatment effect under unconfoundedness, whose influence function $\frac{A}{e(X)}[Y - m_1(X)] + m_1(X) - m_0(X) - \psi_0$ is known to be efficient [21]; the same general framework applies whenever the nuisance is infinite-dimensional.

Definition 4.1 (Regular asymptotic linearity, EIF, efficiency bound). *An estimator $\hat{\theta}$ of a functional $\theta(\mathbb{P})$ is regular asymptotically linear (RAL) if*

$$\hat{\theta} - \theta(\mathbb{P}) = \mathbb{P}_n[\varphi(O)] + o_P(n^{-1/2})$$

for some centred $\varphi \in L^2(\mathbb{P})$ whose limiting distribution is invariant under local perturbations of $\mathbb{P}$ along regular parametric submodels. The admissible influence functions form a Hilbert subspace, the tangent space of the model [22]; its unique variance-minimising element is the efficient influence function (EIF) φ^ . The variance $\mathbb{E}[(\varphi^*)^2]$ is the semiparametric efficiency bound, the smallest achievable among all regular asymptotically unbiased estimators.*

Cui et al. [8] apply this framework to the proximal model.

Applying this framework, Cui et al. [8] project the plug-in score onto the proximal tangent space and obtain an EIF with the structure *plug-in plus residual correction*. The residual is weighted by a new nuisance, the treatment bridge q_0 , defined by a moment equation symmetric to (3.1).

4.2 The proximal EIF and the IPW form

Proposition 4.1 (Proximal EIF for the binary ATE; Cui, Pu, Shi, Miao, and Tchetgen Tchetgen 8, Thm. 3.1). *Under assumptions 2.1 to 2.6, given any solutions h_0 to (3.1) and q_0 to the treatment-bridge moment equation (4.3), the efficient influence function of $\psi_0 = \mathbb{E}[Y(1) - Y(0)]$ takes the form*

$$\varphi^*(O; h_0, q_0) = h_0(W, 1, X) - h_0(W, 0, X) + (-1)^{1-A} q_0(Z, A, X) [Y - h_0(W, A, X)] - \psi_0. \quad (4.2)$$

The structure combines the plug-in contrast with a residual correction $Y - h_0(W, A, X)$ weighted by the treatment bridge q_0 .

Recall that under unconfoundedness, the AIPW influence function weights the residual by the inverse propensity score $1/e(X)$, which transforms a treated observation into an estimate for the full population. The proximal analogue $1/f(A | U, X)$ is unavailable because U is latent. Cui et al. [8] introduce the treatment bridge q_0 as any solution of the conditional moment equation

$$\mathbb{E}[q_0(Z, A, X) | W, A, X] = \frac{1}{f(A | W, X)}, \quad (4.3)$$

which is symmetric to the outcome-bridge equation (3.1), with the conditioning side and the function side exchanged. The right-hand side is observable, so (4.3) is a Fredholm equation in q_0 that the data can solve in principle. A consequence of (4.3), central to the IPW reading below, is that q_0 acts on the latent confounder as the proximal proxy of the inverse propensity score:

$$\mathbb{E}[\pi(A | X) q_0(Z, A, X) | A, U, X] = \frac{\pi(A | X)}{f(A | U, X)}. \quad (4.4)$$

Equivalently, q_0 identifies the causal target through an IPW formula [8]: in the binary case,

$$\mathbb{E}[Y(a)] = \mathbb{E}[(-1)^{1-A} q_0(Z, a, X) Y]. \quad (4.5)$$

Proximal identification therefore proceeds along two complementary routes, the g-formula via h_0 and the IPW formula via q_0 . The EIF in (4.2) combines both into an augmented score.

Existence and uniqueness of q_0 require the symmetric counterpart of Assumption 2.6, on Z given (W, A, X) [8]:

Assumption 4.1 (Completeness of Z ; Cui et al. [8]). For any $g \in L^2$, $\mathbb{E}[g(U) | Z, A, X] = 0$ a.s. implies $g(U) = 0$ a.s.

The pair (h_0, q_0) thus lives under two completeness conditions; neither is empirically testable, by the same continuity argument as in Section 3.1 [18]. In Cui's binary setting q_0 is policy-free; in Kallus's extension to a stochastic policy π it appears in the influence function only in product form $\pi \cdot q_0$.

4.3 The doubly robust estimator and its bias

Kallus et al. [9] carry the binary EIF construction of Section 4.2 over to the continuous setting and the working estimand of this essay, showing that the score

$$\varphi^*(O) = T_\pi h_0(W, X) + \pi(A | X) q_0(Z, A, X) [Y - h_0(W, A, X)] - J(\pi_{a,b}) \quad (4.6)$$

is the efficient influence function of $J(\pi_{a,b})$ under the proximal model. The estimator built from (4.6) is therefore regular and asymptotically unbiased, attaining the semiparametric efficiency bound for the GACE. The binary efficiency result of Cui et al. [8] carries over without loss when both bridges are uniquely identified. The estimator itself, distinct from the plug-in $\hat{J}_{\text{REG}}$ of Section 4.1, is

$$\hat{J}_{\text{DR}} = \mathbb{P}_n [T_\pi \hat{h}(W, X) + \pi(A | X) \hat{q}(Z, A, X) (Y - \hat{h}(W, A, X))]. \quad (4.7)$$

To understand the behaviour of $\hat{J}_{\text{DR}}$, we read off its asymptotic decomposition, which separates a sampling fluctuation from a bridge-induced bias and a higher-order remainder:

$$\hat{J}_{\text{DR}} - J = \underbrace{\left(\frac{1}{n} \sum_{i=1}^n \varphi^*(O_i) - \mathbb{E}[\varphi^*(O)] \right)}_{\text{(a) empirical}} + \underbrace{\left(\mathbb{E}[\varphi^*(O; \hat{h}, \hat{q})] - J \right)}_{\text{(b) bias}} + \underbrace{R_n}_{\text{(c) remainder}}. \quad (4.8)$$

Term (a) is governed by the central limit theorem, so $\sqrt{n}(\frac{1}{n} \sum_{i=1}^n \varphi^*(O_i) - \mathbb{E}[\varphi^*(O)]) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$ with $\sigma^2 = \mathbb{E}[(\varphi^*)^2]$ the semiparametric efficiency bound; term (c) is $o_P(n^{-1/2})$ under standard empirical-process conditions [22], which sample-splitting or cross-fitting enforces by construction. The behaviour of the estimator is therefore controlled by the bias term (b), which the next theorem characterises.

Theorem 4.1 (Double robustness and semiparametric efficiency of $\hat{J}_{\text{DR}}$; Cui, Pu, Shi, Miao, and Tchetgen Tchetgen 8, Thm. 3.2). *Suppose assumptions 2.1 to 2.6 together with Assumption 4.1, that the bridges h_0 and q_0 are uniquely identified, and that the cross-fitted nuisance estimators satisfy $\|\hat{h} - h_0\|_{L^2} = o_P(n^{-1/4})$ and $\|\hat{q} - q_0\|_{L^2} = o_P(n^{-1/4})$. Then*

- (i) $\hat{J}_{\text{DR}}$ is $\sqrt{n}$ -asymptotically normal, $\sqrt{n}(\hat{J}_{\text{DR}} - J) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$, with $\sigma^2 = \mathbb{E}[(\varphi^*)^2]$ equal to the semiparametric efficiency bound;
- (ii) $\hat{J}_{\text{DR}}$ is doubly robust: if either $\hat{h} \rightarrow h_0$ or $\hat{q} \rightarrow q_0$ in L^2 , the estimator is consistent for J even when the other bridge is misspecified.

Proof. Write $\delta_h = \hat{h} - h_0$ and $\delta_q = \hat{q} - q_0$. Substituting $\hat{h} = h_0 + \delta_h$ and $\hat{q} = q_0 + \delta_q$ in $\mathbb{E}[\varphi^*(O; \hat{h}, \hat{q})]$ and expanding, two cancellations occur by construction. The first-order term in δ_h vanishes because q_0 satisfies the dual moment equation that defines it [8]; that in δ_q vanishes because the residual $Y - h_0(W, A, X)$ is conditionally centred on (Z, A, X) by the bridge equation (3.1). Only the cross term survives:

$$\mathbb{E}[\varphi^*(O; \hat{h}, \hat{q})] - J = -\mathbb{E}[\pi(A | X) \delta_q(Z, A, X) \delta_h(W, A, X)] + o(\|\delta_h\| \|\delta_q\|). \quad (4.9)$$

By Cauchy–Schwarz,

$$|\text{Bias}| \leq C \|\delta_h\|_{L^2} \|\delta_q\|_{L^2}. \quad (4.10)$$

Under the mixed-rate condition, $\|\delta_h\|_{L^2} \|\delta_q\|_{L^2} = o_P(n^{-1/2})$, hence $\sqrt{n} |\text{Bias}| = o_P(1)$ and term (b) of (4.8) is asymptotically negligible; combined with the CLT for term (a) and the empirical-process control of term (c), this gives (i). For (ii), if $\delta_h = 0$ or $\delta_q = 0$ exactly, the right-hand side of (4.9) vanishes and consistency follows from (a) and (c) alone, regardless of the convergence of the other bridge. $\square$

The product structure of (4.9) reflects the orthogonality of the canonical score: double robustness emerges as a consequence of the construction.

Theorem 4.1 delivers the optimal guarantee of semiparametric theory, at the cost of three demanding requirements: two non-testable completeness conditions, the unique identification of both bridges, and a uniform L^2 -control of the estimation errors at rate $n^{-1/4}$, which makes no distinction between directions that do and directions that do not contribute to the functional of interest. The brittleness of these requirements in ill-posed problems motivates the next development, where the uniqueness of the bridges is dropped in favour of a weaker identification condition on the functional itself.

4.4 Strong identification of the functional

Bennett et al. [10] retain the estimator structure of (4.7) but relax the uniqueness requirement on h_0 : the functional $J(\pi_{a,b})$ can be estimated at the $\sqrt{n}$ -rate provided it is *strongly identified* in the sense defined below, even when distinct functions solve the bridge equation (3.1). The control shifts from the bridge in $L^2(W, A, X)$ to its image under T in $L^2(Z, A, X)$. This projected metric is insensitive to the kernel of T .

The Riesz representation theorem states that every continuous (i.e. bounded) linear form on a Hilbert space $\mathcal{H}$ is the inner product with a unique element of $\mathcal{H}$. The map $h_0 \mapsto J(h_0) = \mathbb{E}[T_\pi h_0]$ is linear and continuous on $L^2(W, A, X)$, the latter by boundedness of T_π . The theorem applies: there is a unique $\alpha_\pi \in L^2(W, A, X)$ such that

$$J(h_0) = \langle \alpha_\pi, h_0 \rangle_{L^2(W, A, X)}. \quad (4.11)$$

The existence of α_π guarantees a representation of the functional. Stability of the debiasing weight, however, requires further structure.

A debiasing weight $r \in L^2(Z, A, X)$ kills the first-order sensitivity of the score whenever it satisfies the dual equation

$$T^*r = \alpha_\pi. \quad (4.12)$$

Equation (4.12) typically admits many solutions, and not every choice is equally well-behaved. Applying the Riesz identity (4.11) to the g-formula term and the bridge equation (3.1) to the correction term, then substituting $\alpha_\pi = T^*r$ from (4.12),

$$\begin{aligned} \mathbb{E}[\varphi^*(O; \hat{h}, \hat{r})] - J(h_0) &= \langle \alpha_\pi - T^*\hat{r}, \hat{h} - h_0 \rangle_{L^2(W, A, X)} \\ &= -\langle T^*(\hat{r} - r), \hat{h} - h_0 \rangle_{L^2(W, A, X)}. \end{aligned} \quad (4.13)$$

The error $T^*(\hat{r} - r)$ maps the residual from $L^2(Z, A, X)$ into $L^2(W, A, X)$; no single projected metric controls both factors simultaneously, and the bias is sensitive to the ill-posedness of T^* . Bennett et al. [10] resolve this by restricting the debiasing weight to the family

$$r = T\xi, \quad \xi \in L^2(W, A, X), \quad (4.14)$$

which puts ξ in the same space as the bridge. Substituting (4.14) into the dual equation (4.12) yields the *source condition*

$$T^*T\xi_0 = \alpha_\pi, \quad (4.15)$$

which requires α_π to lie in the range of T^*T . The functional J is said to be *strongly identified* [10] when (4.15) admits a solution ξ_0 ; the dual weight in the score is then $r_\pi = T\xi_0$.

Specialising to $\hat{r} = T\xi$ and $r = r_\pi = T\xi_0$ in (4.13) and using the adjoint identity $\langle T^*f, g \rangle_{L^2(W, A, X)} = \langle f, Tg \rangle_{L^2(Z, A, X)}$ gives

$$\mathbb{E}[\varphi^*(O; h, T\xi)] - J(h_0) = -\langle T(\xi - \xi_0), T(h - h_0) \rangle_{L^2(Z, A, X)}, \quad (4.16)$$

and Cauchy–Schwarz yields

$$|\text{Bias}| \leq \|T(\hat{\xi} - \xi_0)\|_{L^2(Z, A, X)} \cdot \|T(\hat{h} - h_0)\|_{L^2(Z, A, X)}. \quad (4.17)$$

Both norms in (4.17) measure the bridge error *seen through* T : $\|T(\hat{h} - h_0)\|_{L^2(Z, A, X)}$ is the size of the conditional residual $\mathbb{E}[\hat{h} - h_0 \mid Z, A, X]$, which conditional projection strictly contracts whenever $\hat{h} - h_0$ has a component in the kernel of T . The control is therefore on the moment equation in projection rather than on bridge recovery in $L^2(W, A, X)$, and is insensitive to ill-posedness; Bennett trades the uniqueness of h_0 for projected control of the bias. Under the decomposition of (4.8), terms (a) and (c) are handled exactly as in the Cui–Kallus argument; the rate condition controls term (b) without requiring h_0 to be unique. Bennett et al. [10] establish $\sqrt{n}$ -asymptotic normality under strong identification together with the rate condition $\|T\delta_h\| \|T\delta_\xi\| = o_P(n^{-1/2})$ and consistency in the strong norm $o_P(1)$. The source condition (4.15) is the Euler–Lagrange equation of the variational problem

$$\xi_0 = \arg \min_{\xi \in L^2(W, A, X)} \left\{ \frac{1}{2} \|T\xi\|_{L^2(Z, A, X)}^2 - J(\xi) \right\}, \quad (4.18)$$

which estimation algorithms solve in practice.

4.5 The open question of efficiency

No current theory establishes efficiency under Bennett's source condition; this remains an open question. The three layers of this section differ in their statistical guarantees. The plug-in only inherits the nonparametric bridge rate. Cross-fitted DR adds asymptotic normality at the efficiency bound, provided both bridges are uniquely identified. Bennett's source condition relaxes the uniqueness requirement but no longer guarantees efficiency. The source condition is weaker than the orthogonality that Cui and Kallus exploit, so confidence intervals from strongly identified estimators are asymptotically valid but need not be of minimal width.

Section 5 develops the spectral conditions on the bridge estimation problem and connects them to the tuning criteria for the estimators of Section 6.

5 Ill-Posedness and Functional Regularity

Section 4.5 closed on an open question that Bennett's framework leaves: $\sqrt{n}$ -asymptotic normality holds under the source condition, but the stability of the resulting inference depends on a notion of *regularity of the Riesz representer* relative to the bridge operator T that neither Cui–Kallus nor Bennett quantify completely. How is this regularity measured, and what lever can the analyst pull to improve it without altering the causal target? A spectral reading makes the regularity precise through a continuous parameter β indexing a hierarchy of source conditions; the noisy binary treatment then provides the applied case where the framework applies cleanly, with smoothing in the treatment argument lifting the hard contrast off the weakly identified directions of the proximal inversion.

5.1 The source condition

The source condition of Section 4.4, $T^*T\xi_0 = \alpha_\pi$, is most naturally read in spectral terms. Throughout this section we assume that $K_T = T^*T$ is a compact, self-adjoint, positive operator on $L^2(W, A, X)$, with spectral decomposition $K_T e_j = s_j^2 e_j$ and singular values $s_j \rightarrow 0$ as $j \rightarrow \infty$. The decay of s_j is the spectral signature of the ill-posedness of the Fredholm equation (3.1): small s_j correspond to *weakly identified directions* of the bridge. This setting is standard for conditional expectation operators on RKHS [10].

It will be convenient to decompose the Riesz representer in the spectral basis: $\alpha_\pi = \sum_j a_j(\pi) e_j$. The source condition of Section 4.4 required $\alpha_\pi \in \text{Range}(K_T)$, which is the case $\beta = 1$ of the broader source condition of degree β considered by Bennett et al. [10]. The latter extends to arbitrary $\beta > 0$ through membership in $\text{Range}(K_T^{\beta/2})$, equivalently the existence of $g \in L^2$ with $\alpha_\pi = K_T^{\beta/2} g$, whence $g = K_T^{-\beta/2} \alpha_\pi$ is square-integrable. In the spectral basis, $K_T^{-\beta/2} e_j = s_j^{-\beta} e_j$, so that

$$\|K_T^{-\beta/2} \alpha_\pi\|^2 = \mathcal{I}_\beta(\pi) = \sum_j \frac{|a_j(\pi)|^2}{s_j^{2\beta}}, \quad (5.1)$$

and the regularity condition reads $\mathcal{I}_\beta(\pi) < \infty$. The degree β is the maximal order of regularity of the representer relative to K_T ; the *source index* $\mathcal{I}_\beta$ is its norm in that order. The functional is stable to estimate when α_π concentrates on directions with s_j bounded away from zero, and becomes unstable as soon as the representer loads the small singular values.

The source index is not the only available measure of ill-posedness; two others occur in the literature, and the contrast with them clarifies what $\mathcal{I}_\beta$ adds.

- (i) The polynomial decay rate $s_j \asymp j^{-r}$ of the singular values is the most intrinsic: it depends on the operator alone and bounds how ill-posed the bridge equation can be irrespective of any functional one wishes to estimate.
- (ii) The conditioning of the inverse T^{-1} , used by Kallus et al. [9] in the convergence analysis of Theorem 7 for the outcome bridge h_0 , also depends on the operator alone, but as a global norm rather than a spectral rate.

The source index $\mathcal{I}_\beta$ differs from both: it measures the regularity of the *pair* (α_π, K_T) and is therefore *policy-aware*. Because $\mathcal{I}_\beta$ depends on π , and through π on the bandwidth b of Section 2.3, it depends on choices the analyst can make. Section 5.2 examines whether those choices amount to a genuine lever on stability, or whether β is a fixed feature of the problem.

5.2 Smoothing in A and functional stability

We propose to read the bandwidth b of the kernel policy $\pi_{a,b}$ defined in Section 2.3 in two ways: as the target-defining parameter of Equation (2.4) and as the analyst's only lever on functional stability. The lever is partial by construction. The kernel policy smooths in argument of treatment A alone, whereas the Riesz representer α_π lives on the full tuple (W, A, X) ; whatever the inversion identifies weakly through (W, X) remains unaffected by any choice of b . The complementary part is where the lever bites. Among the directions that T identifies weakly, those associated with rapid variations of the representer in A are reachable through the treatment coordinate; raising b shifts the weight of α_π away from them and stabilises the inversion.

Section 5.3 works through the noisy binary treatment, the case where this lever can be analysed rigorously and where the gain in stability admits an explicit closed form.

5.3 The noisy binary treatment as a stochastic policy

Many treatments recorded as binary cover, in reality, distributed interventions, as anticipated in Section 2.3. The influenza vaccination of Section 1 aggregates vaccine variants, time of administration, and immune response within a single recorded value; a pharmaceutical compound varies in compliance and absorbed dose; an environmental exposure varies in duration and intensity. The hard binary contrast $\mathbb{E}[Y(1) - Y(0)]$ is then an idealisation; the actual causal target is a contrast of distributions of latent continuous exposures, $\Pi_{1,\rho} - \Pi_{0,\rho}$, conditional on the recorded binary value. Throughout this subsection ρ denotes the dispersion of the analyst's stochastic policy (ρ_{pol} in the simulation notation of Section 7); the unknown latent dispersion of the true intervention is ρ_{true} . The two coincide at the calibration optimum identified empirically in Section 7.

The honest causal estimand is

$$J_\rho = \mathbb{E} \left[\int Y(t) (\Pi_{1,\rho} - \Pi_{0,\rho})(dt) \right], \quad (5.2)$$

a special case of the GACE of Kallus et al. [9] extended to a contrast of stochastic policies on a latent continuous exposure.

Replacing the hard contrast by the stochastic one moves the target, and a second-order expansion of m around μ_b quantifies the shift.

Proposition 5.1 (Causal cost of the stochastic policy at order $O(\rho^2)$). *Suppose $\Pi_{b,\rho} = \mathcal{N}(\mu_b, \rho^2)$ for $b \in \{0, 1\}$ and $m \in \mathcal{C}^2$. Then*

$$J_\rho = (m(\mu_1) - m(\mu_0)) + \frac{\rho^2}{2} (m''(\mu_1) - m''(\mu_0)) + o(\rho^2).$$

Sketch. For $T_b \sim \mathcal{N}(\mu_b, \rho^2)$, a Taylor expansion of m around μ_b to second order combined with $\mathbb{E}[T_b - \mu_b] = 0$ and $\mathbb{E}[(T_b - \mu_b)^2] = \rho^2$ gives $\mathbb{E}[m(T_b)] = m(\mu_b) + \frac{\rho^2}{2} m''(\mu_b) + o(\rho^2)$. Differencing over b yields the claim. $\square$

The estimand therefore moves at second order in the local curvature of the dose-response, opening a tradeoff between bias from over-smoothing and stability gain. We see this tradeoff empirically in the U-curve traced by the DGP3 experiment of Section 7.

The compensating benefit of the stochastic relaxation is a strict gain of functional stability that admits an explicit spectral form, read off from the Fourier representation of the smoothing operator induced by the policy.

The Gaussian policy of dispersion ρ on the latent continuous exposure acts on integrable functions of $\mathcal{A}$ by convolution with the Gaussian kernel K_ρ ,

$$(S_\rho f)(t) = \int K_\rho(t-s) f(s) ds = (K_\rho * f)(t). \quad (5.3)$$

The convolution theorem maps this convolution to a multiplication in frequency, $\widehat{S_\rho f}(\omega) = \widehat{K_\rho}(\omega) \widehat{f}(\omega)$, with $\widehat{K_\rho}(\omega) = e^{-\rho^2 \omega^2 / 2}$ for the Gaussian kernel. The Fourier modes $e_j(t) = e^{i\omega_j t}$ are therefore eigenfunctions of S_ρ ,

$$S_\rho e_j = e^{-\rho^2 \omega_j^2 / 2} e_j. \quad (5.4)$$

Bochner's theorem ensures that this Fourier diagonalisation is canonical for any positive-definite translation-invariant kernel: such a kernel admits a representation $\kappa(t) = \int e^{i\omega t} d\Lambda(\omega)$ for a positive spectral measure Λ , itself Gaussian for the Gaussian kernel.³

The closed form for $\mathcal{I}_\beta$ requires that this Fourier basis diagonalise both S_ρ and the proximal operator. We adopt the codiagonalisation hypothesis

$$K_T e_j = s_j^2 e_j, \quad S_\rho e_j = e^{-\rho^2 \omega_j^2 / 2} e_j, \quad (5.5)$$

under which the high-frequency Fourier modes coincide with the weakly identified directions of T . The first relation is the spectral assumption of Section 5.1 specialised to the Fourier basis; the second is (5.4).

Decomposing the hard binary contrast in this basis as $\delta_{\mu_1} - \delta_{\mu_0} = \sum_j a_j(0) e_j$, the stochastic contrast inherits the eigenvalues of S_ρ ,

$$\Pi_{1,\rho} - \Pi_{0,\rho} = S_\rho(\delta_{\mu_1} - \delta_{\mu_0}) = \sum_j a_j(0) e^{-\rho^2 \omega_j^2 / 2} e_j.$$

Substituting into (5.1) and using the codiagonalisation (5.5) to read $K_T^{-\beta/2} e_j = s_j^{-\beta} e_j$ in the same basis gives the closed form

$$\mathcal{I}_\beta(\Pi_{1,\rho} - \Pi_{0,\rho}) = \sum_j \frac{|a_j(0)|^2 e^{-\rho^2 \omega_j^2}}{s_j^{2\beta}} \asymp \sum_j |a_j(0)|^2 e^{-\rho^2 \omega_j^2} \omega_j^{2r\beta}, \quad (5.6)$$

the last equivalence under the polynomial decay $s_j \asymp \omega_j^{-r}$. The Gaussian factor $e^{-\rho^2 \omega_j^2}$ attenuates exactly the high-frequency modes whose polynomial growth $\omega_j^{2r\beta}$ would otherwise drive the sum to diverge. Whether the divergence at $\rho = 0$ is genuine or merely apparent depends on how the hard contrast loads the high frequencies.

Proposition 5.2 (Stability gain at the singular limit). *Suppose K_T has polynomially decaying singular values $s_j \asymp j^{-r}$, and the binary contrast spectral coefficients $|a_j(0)|^2$ do not decay with j . Then for any $\beta > 0$,*

$$\mathcal{I}_\beta(\pi_{1,0} - \pi_{0,0}) = \infty, \quad \mathcal{I}_\beta(\Pi_{1,\rho} - \Pi_{0,\rho}) < \infty \quad \forall \rho > 0.$$

The non-decay assumption on $|a_j(0)|^2$ is structural: the Fourier transform of $\delta_{\mu_1} - \delta_{\mu_0}$ is $e^{-i\omega\mu_1} - e^{-i\omega\mu_0}$, of modulus $|2 \sin(\omega(\mu_1 - \mu_0)/2)|$, which oscillates in $[0, 2]$ without converging to zero. The hard contrast is intrinsically loaded with high frequencies, which is what makes $\rho = 0$ a singular point of $\rho \mapsto \mathcal{I}_\beta(\rho)$.

³Bochner's theorem characterises positive-definite translation-invariant kernels through a spectral measure; Mercer's theorem decomposes a kernel into eigenfunctions of an integral operator on a compact domain. The Fourier basis is the natural spectral basis of S_ρ but does not generically diagonalise the proximal operator $K_T = T^*T$, hence the codiagonalisation hypothesis stated as (5.5) below. The hypothesis holds in the proximal RKHS-translation-invariant setting whose simulations illustrate this section, but is not generic and would fail under non-translation-invariant proxy structure. A natural direction for further inquiry is the behaviour of $\mathcal{I}_\beta$ when codiagonalisation fails and the high-frequency Fourier modes are merely correlated with the weakly identified directions of T rather than aligned with them.

Sketch. At $\rho = 0$, (5.6) reduces to $\sum_j |a_j(0)|^2 / s_j^{2\beta} \asymp \sum_j j^{2r\beta}$, which diverges since $|a_j(0)|^2$ does not decay. For $\rho > 0$, the Gaussian factor $e^{-\rho^2 \omega_j^2}$ dominates any polynomial in ω_j , so each term of (5.6) is summable. $\square$

5.4 Synthesis

Section 5.3 replaces the idealised binary contrast by its actual stochastic counterpart and turns the policy dispersion into a spectral lever loosening the source condition that Bennett's framework requires. We calibrate ρ against the U-curve of Section 7 as a domain-expertise input rather than tuning it as a free statistical parameter. We turn next to Section 6, which translates the doubly robust estimators of Section 4 into the RKHS implementation in which Fourier diagonalisation and the source index acquire numerical counterparts.

6 Numerical Implementation

The estimators of Sections 3 and 4, refined through the spectral analysis of Section 5, need a numerical realisation⁴ before they can be applied to data. Three questions shape that realisation: how to solve the Fredholm inversions underlying the bridges in finite samples, which feature approximations reduce the cost to a tractable order without distorting the inversion, and which empirical quantities tell the analyst whether the resulting estimator is well-calibrated.

Section 6.1 collects the numerical tools shared across methods, then Section 6.2 derives the finite system that each of the four estimators solves; Section 6.3 closes with the calibration and diagnostic metrics that the next section uses for tuning and reporting.

6.1 Shared numerical tools

The four methods of Section 6.2 share a small toolkit of numerical techniques presented below.

Numerical safeguards. The ridge Grams $G + n\lambda I$ are symmetric positive-definite in theory but can fail this property numerically through rounding asymmetry or near-zero eigenvalues. Three ingredients preserve the Cholesky factorisation: an explicit symmetrisation $G \leftarrow (G + G^\top)/2$ before Cholesky; a jitter retry $G + \varepsilon \|G\| I$ with $\varepsilon = 10^{-8}$ when the first attempt fails; and a fallback chain Cholesky $\rightarrow$ SPD solver $\rightarrow$ regularised least squares for the residual cases. On the IPW correction $K_h(A - a) \hat{q}(Z, A) (Y - \hat{h}(W, A))$, the variance scales with the second moment of the weight $\hat{q}$: a handful of extreme positive values in low-density regions of $A \mid Z$ then dominates that moment and inflates the variance of the score. We cap weights above $M_n = 5 n^{1/4}$ to M_n but leave moderate and negative values untouched: extreme positive excursions reflect low-density regions rather than the bridge structure and would carry disproportionate weight, whereas negative excursions are the cancellation channel of the doubly robust mechanism.⁵

Calibration. The residual $Y_i - \hat{h}(W_i, A_i)$ is not independent of $\hat{h}$ when observation i has been used to fit it, and this self-observation invalidates the asymptotic guarantees of the doubly robust score. The K-fold cross-fitted evaluation procedure of Chernozhukov et al. [24], applied to the proximal setting by Ying and Tchetgen [25], circumvents the issue by partitioning the sample into K disjoint folds: each bridge ($\hat{h}$ as well as $\hat{q}$) is trained on $K - 1$ folds and evaluated on the held-out fold, with a fresh model instance per fold to prevent state leakage. The aggregated score is then built exclusively on (bridge, observation) pairs that are independent. We retain $K = 5$ throughout, the standard choice in cross-fitted DR [24]. Bandwidths

⁴The implementation patterns in this section draw on the companion code repositories of Bennett et al. [10] (CausalML/Inverse-Problems-Debiased-Inference and syrgkanislab/npiv_functionals) and of Wu et al. [23] (PKU-ML/PCL.Continuous.Treatment). We adapt the dual minimax solve, the K-fold cross-fitting harness, and the two-stage kernel-ridge structure, replacing neural-net function classes with random-Fourier-feature parameterisations and deriving closed-form policy moments under a Gaussian kernel policy.

⁵The $n^{1/4}$ scaling matches the asymptotic growth rate of $\hat{q}$ under standard ill-posedness. We picked the factor 5 by trial-and-error on the pilot runs of Section 7: it rarely activates in regular cells but still bounds variance in heavy-tailed ones.

start from two deterministic heuristics, each refined empirically per method before any production run. The median heuristic $\ell = \text{median}\{\|x_i - x_j\| : i < j\}$ initialises RBF length-scales on the empirical pairwise distance distribution; the kernel-policy bandwidth starts from Silverman's rule $b_{\text{KDE}} = 1.06 \hat{\sigma}(A) n^{-1/5}$ [26]. Neither heuristic depends on the random seed, which protects calibration against overfitting to a particular Monte-Carlo batch.

Numerical optimisation. The dominant cost of an RKHS bridge solve is the $O(n^3)$ Cholesky on the $n \times n$ Gram. Two finite-feature approximations reduce this to $O(nm^2)$ with $m \ll n$. The Nyström approximation $\hat{K} = K_m K_{mm}^{-1} K_m$ projects onto m uniformly sampled landmarks [27]; Random Fourier Features instead exploit Bochner's theorem to identify the spectral decomposition of any translation-invariant kernel with its Fourier transform, sampling m frequencies $\omega_j \sim \text{FT}(k)$ to produce a feature map $\Phi(x) = \sqrt{2/m}(\cos(\omega_j^\top x + b_j))_{j=1}^m$ such that $\mathbb{E}[\Phi(x)^\top \Phi(y)] = k(x, y)$ [28]. A separate optimisation eliminates numerical quadrature on the policy side: for $\pi_{a,b}(t) = K_h(t - a)$, the integral $\int \hat{h}(W, t) K_h(t - a) dt$ admits a closed form under both bridge constructions. With a KPV bridge built on an RBF kernel of length-scale ℓ_A on the treatment, the convolution of two Gaussians yields the moment

$$\tilde{k}(A_j, a, b) = \frac{\ell_A}{\sqrt{\ell_A^2 + b^2}} \exp\left(-\frac{(A_j - a)^2}{2(\ell_A^2 + b^2)}\right), \quad (6.1)$$

evaluated at each observation's A_j . With a Bennett bridge built on RFF features, the action of the policy on a cosine feature is the spectral damping $\mathbb{E}_{A \sim \mathcal{N}(a, b^2)}[\cos(\omega_W W + \omega_A A + b)] = \cos(\omega_W W + \omega_A a + b) \exp(-\omega_A^2 b^2 / 2)$, the same Gaussian factor that appeared in (5.6).

6.2 Analytical derivations of the four methods

The four methods compared in Section 7 are: (i) a linear two-stage with generated regressor, (ii) the KPV plug-in, (iii) a doubly robust score with cross-fit (DRKPV), and (iv) the Bennett score with RFF moment-solve bridges.

6.2.1 Linear two-stage with generated regressor

First stage, project W on the instruments:

$$(X_W^\top X_W) \hat{\gamma} = X_W^\top W, \quad \hat{W} = X_W \hat{\gamma}, \quad X_W = [\mathbf{1}, A, X, Z]. \quad (6.2)$$

Second stage, regress Y on $(A, X, \hat{W})$:

$$(S_2^\top S_2) \hat{\beta} = S_2^\top Y, \quad S_2 = [\mathbf{1}, A, X, \hat{W}], \quad (6.3)$$

$\hat{\beta}_A$ being the coefficient on A in the second-stage design S_2 .

Note that the textbook OLS formula $\sigma^2(S_2^\top S_2)^{-1}$ underestimates the variance of $\hat{\beta}_A$ because it treats $\hat{W}$ as observed rather than as a noisy estimate inheriting the sampling fluctuation of $\hat{\gamma}$. Pagan [29] provides the standard sandwich correction that propagates first-stage uncertainty into the second-stage variance, and we use it throughout.

6.2.2 KPV plug-in: two-stage RKHS for the outcome bridge

We follow the two-stage RKHS formulation of Mastouri et al. [20], which avoids inverting the Fredholm operator explicitly by writing the unknown bridge as a function of an estimated conditional mean embedding rather than as the solution of an integral equation.

First stage: conditional mean embedding. For an RKHS $\mathcal{H}_W$ on the proxy with kernel k_W , the mean embedding $\mu_{W|Z=z, A=a} \in \mathcal{H}_W$ admits the quadratic risk $\mathbb{E}[\|k_W(W, \cdot) - \mu_{W|Z, A}\|_{\mathcal{H}_W}^2]$ whose minimiser is the conditional mean embedding. Working in a second tensor-product RKHS on (Z, A) with kernel

$k_{ZA}((z, a), (z', a')) = k_Z(z, z') k_A(a, a')$ and applying the representer theorem to the regularised empirical risk

$$\min_{\mu \in \mathcal{H}_W \otimes \mathcal{H}_{ZA}} \frac{1}{n} \sum_{i=1}^n \|k_W(W_i, \cdot) - \mu(Z_i, A_i)\|_{\mathcal{H}_W}^2 + \lambda_1 \|\mu\|^2,$$

turns the infinite-dimensional problem into a finite linear system in n coefficients per training point, and solving the normal equations yields

$$\Gamma = (K_{ZA} + n\lambda_1 I)^{-1} K_{ZA} \in \mathbb{R}^{n \times n}, \quad K_{ZA} = K_Z \odot K_A, \quad (6.4)$$

with K_W, K_Z, K_A the empirical Grams on the training sample and the Hadamard product encoding the tensor structure on (Z, A) . The matrix Γ is the coefficient matrix of the conditional mean embedding itself: at any query (z, a) , $\hat{\mu}_{W|Z=z, A=a} = \sum_i (\Gamma k_{ZA}(\cdot, (z, a)))_i k_W(W_i, \cdot)$, so each column of Γ collects the n weights placed on the training proxies to reconstruct the posterior over W at one training point of (Z, A) .

Second stage: ridge on tensor features. With the embedding $\hat{\mu}_i$ in hand, the bridge ansatz combines the embedded proxy representation $\hat{\mu}(w, a) \in \mathcal{H}_W$ with a treatment feature $\varphi_A(a) \in \mathcal{H}_A$ via tensor product: $h(w, a) = \langle \hat{\mu}(w, a) \otimes \varphi_A(a), \beta \rangle$ in the joint RKHS. The empirical risk

$$\min_h \frac{1}{n} \sum_{i=1}^n (Y_i - h(W_i, A_i))^2 + \lambda_2 \|h\|^2$$

again reduces to a finite ridge by the representer theorem. The factorisation (6.4) allows the associated Gram matrix to be written as $G = (\Gamma^\top K_W \Gamma) \odot K_A$, and the normal equations give

$$(G + n\lambda_2 I) \hat{\alpha} = Y. \quad (6.5)$$

The presence of Γ in the middle of G , rather than a plain collocation inverse K_{ZA}^{-1} , distinguishes KPV from a naive Fredholm solver: the embedding step transports the bridge through the proxy RKHS before the ridge fits the outcome [20].

The estimand $J(\pi_{a,b})$ is then evaluated without any quadrature through the Gauss–Gauss convolution (6.1): the moment of the policy kernel K_h against k_A has a closed form, and the plug-in estimand

$$\hat{J}_{\text{REG}}(\pi_{a,b}) = \frac{1}{n} \sum_{i,j} \hat{\alpha}_j (K_W(W_i, \cdot) \Gamma)_j \tilde{k}(A_j, a, b)$$

reduces to matrix products only.

6.2.3 Doubly robust score with cross-fit

The doubly robust Version B score of Cui et al. [8], Kallus et al. [9] complements the plug-in of (6.4)–(6.5) with an IPW-style residual correction:

$$\hat{J}_{\text{DR}}(\pi_{a,b}) = \frac{1}{n} \sum_i \left[K_h(A_i - a) \hat{q}_a(Z_i, A_i) (Y_i - \hat{h}(W_i, A_i)) + \int \hat{h}(W_i, t) K_h(t - a) dt \right], \quad (6.6)$$

with $\hat{h}$ refit per fold and $\hat{q}_a$ a policy-specific treatment bridge defined dose by dose.

Policy-weighted moment. For each target dose a , Kallus et al. [9] seek the population bridge $q_{0,a}$ satisfying

$$\mathbb{E}[\pi_{a,b}(A) q_{0,a}(Z, A) g(W, A)] = \mathbb{E}[T_{\pi_{a,b}} g(W)] \quad \forall g \in \mathcal{H}_W \otimes \mathcal{H}_A,$$

the moment that pairs the IPW correction in (6.6) with the plug-in. This is the policy-weighted analogue of the bridge moment behind the PMMR formulation of Section 3.3.2: a full minimax solve would seek the supremum of the moment violation over the test class. We instead take a tensor RKHS ansatz $\hat{q}_a(z, a') =$

$\sum_{\ell} \alpha_{\ell,a} k_Z(Z_{\ell}, z) k_A(A_{\ell}, a')$ and project both sides on the test functions $g_j(w, a') = k_W(W_j, w) k_A(A_j, a')$, reducing the moment to a finite linear system $M_a \alpha_a = b_a$. A Tikhonov penalty $\lambda_Q \|\alpha_a\|^2$ stabilises the inversion and gives the per-dose ridge solution

$$(M_a^{\top} M_a + \lambda_Q I) \hat{\alpha}_a = M_a^{\top} b_a, \quad (6.7)$$

$$M_a = \frac{K_{WA}^{\top} \Lambda_a K_{ZA}}{n}, \quad b_a = \frac{\tilde{k}_a \odot (K_W \mathbf{1}_n)}{n}, \quad \Lambda_a = \text{diag}(K_h(A_i - a)), \quad (6.8)$$

with $K_{WA} = K_W \odot K_A$ and $K_{ZA} = K_Z \odot K_A$ the Hadamard tensor Grams, and $\tilde{k}_a$ the closed-form Gauss–Gauss convolution (6.1) evaluated against each training A_j . This simplification keeps the $\hat{q}_a$ correction tractable but sacrifices the worst-case richness that PMMR would deliver: Section 7 reports configurations in which the resulting DRKPV is dominated by the plug-in KPVREG, and the explanation is structural rather than a matter of tuning.

Double robustness at fixed b . The pairing $r_{\pi_{a,b}}(Z, A) = K_h(A - a) q_{0,a}(Z, A)$ is the Riesz representer of $J(\pi_{a,b})$ on the proximal model: localisation by K_h around the target dose, weighting by the policy bridge $q_{0,a}$. With this identification, the bias of $\hat{J}_{\text{DR}}$ at fixed b takes the cross-product form

$$\text{Bias} = -\mathbb{E}[K_h(A - a) (\hat{q}_a - q_{0,a}) (\hat{h} - h_0)]$$

and vanishes as soon as either factor does.

6.2.4 Bennett score with RFF moment-solve bridges

The target is $J(\pi_{a,b})$ via the Riesz score of Bennett et al. [10], which couples the bridge $\hat{h}$ to a *Riesz representer* $\hat{r}_{\pi_{a,b}}$ tied to the functional itself:

$$\hat{J}_{\text{B}}(\pi_{a,b}) = \frac{1}{n} \sum_i [T_{\pi_{a,b}} \hat{h}(W_i) + \hat{r}_{\pi_{a,b}}(Z_i, A_i) (Y_i - \hat{h}(W_i, A_i))]. \quad (6.9)$$

The two unknowns $\hat{h}$ and $\hat{r}$ are solved by independent moment-solves on the same RFF basis, each starting from a population condition that vanishes in expectation against every test function in the critic span and reducing empirically to a finite linear system. The construction uses three RFF realisations: Φ_H on (W, A) for the bridge ansatz, Φ_C on (Z, A) for the bridge critic, and Ψ on (W, A) for the Riesz test functions.

Bridge $\hat{h}$. For ansatz $h(W, A) = \Phi_H(W, A)^{\top} \beta$ on RFF features and critic $g(Z, A) = \Phi_C(Z, A)^{\top} \theta$, the population condition reads $\mathbb{E}[g(Z, A) (Y - h(W, A))] = 0$ for every g in the critic span. Substituting the ansatz turns this into $\mathbb{E}[\Phi_C(Z, A) (Y - \Phi_H(W, A)^{\top} \beta)] = 0$, whose empirical version is $A_h \beta = d_h$ with $A_h = n^{-1} \Phi_C^{\top} \Phi_H$ and $d_h = n^{-1} \Phi_C^{\top} Y$. The system is overdetermined and the choice of metric on the critic matters: weighting different critics equally treats collinear and orthogonal directions on the same footing. Following the closed-form treatment of linear critic classes in Kallus et al. [9, App. F.2], we solve via the weighted-GMM ridge

$$(A_h^{\top} S_C^{-1} A_h + \lambda_h I) \hat{\beta} = A_h^{\top} S_C^{-1} d_h, \quad S_C = \frac{\Phi_C^{\top} \Phi_C}{n} + \gamma_{\text{critic}} I, \quad (6.10)$$

where S_C^{-1} rescales each critic direction by its inverse empirical covariance.

Riesz $\hat{r}_{\pi_{a,b}}$. For ansatz $r(Z, A) = \Phi(Z, A)^{\top} \gamma$ and test functions $\Psi(W, A)$ spanning the same critic class, the moment condition of Bennett et al. [10] reads $\mathbb{E}[r(Z, A) g(W, A) - T_{\pi_{a,b}} g(W)] = 0$ for every g in the critic span: the Riesz on the proxy side must reproduce the action of the policy operator $T_{\pi_{a,b}}$ on the outcome side, tested against every direction of the critic class. This reduces empirically to $C \gamma = b_a$, with $C = n^{-1} \Psi^{\top} \Phi$ and $b_a = n^{-1} \sum_i \mathbb{E}_{A \sim \pi_{a,b}} [\Psi(W_i, A)]$, the expectation in b_a retaining its closed form under the RFF basis through the spectral damping $\exp(-\omega_A^2 b^2/2)$ of Section 6.1. The ridge solution is

$$(C^{\top} C + \lambda_r I) \hat{\gamma}_a = C^{\top} b_a. \quad (6.11)$$

No two-stage structure is required: the RFF basis is shared between $\hat{h}$ and $\hat{r}$.

Notably, the RFF cosine basis used here is the same Fourier basis that carried (5.6) in the proof of Proposition 5.2, and the diagnostics $\kappa_h, r_{\text{eff},h}, \text{res}_h$ introduced in Section 6.3 are its empirical analogues of the source index $\mathcal{I}_\beta$.

6.3 Calibration and diagnostic metrics

Calibration and diagnosis rest on two families of empirical metrics. The bridge-side family probes the moment-solves themselves: their conditioning, their effective rank, and the residual produced by the linear solver. Score-side metrics complement this by reporting how well the score built on top of the bridges matches its asymptotic prediction, how much effective mass the IPW weights retain, and whether the resulting confidence intervals reach their nominal coverage.

Bridge-side diagnostics. Every moment-solve in Section 6.2 ends in a system of the form $(G + \lambda I)x = b$ with G symmetric positive semi-definite, and three scalars summarise its health:

$$\kappa = \frac{s_{\max}(G + \lambda I)}{s_{\min}(G + \lambda I)}, \quad r_{\text{eff}} = \sum_j \frac{s_j^2}{s_j^2 + \lambda}, \quad \text{res} = \frac{\|G\hat{x} - b\|}{\|b\|}. \quad (6.12)$$

The condition number κ , tracked separately for the $\hat{h}$ -bridge and for the Riesz, is bounded as $n \rightarrow \infty$ under the source condition of Section 5.1; a κ that grows quickly with n signals either a mis-specified bridge class or an insufficient ridge λ . The effective rank r_{eff} counts each singular direction by its identified fraction (close to 1 when $s_j^2 \gg \lambda$, close to 0 when $s_j^2 \ll \lambda$): the sum is the number of directions the regularised system actually learns, and a low r_{eff} corresponds to a regime where the source index $\mathcal{I}_\beta$ is large and the functional is unstable to estimate. The relative residual res should be well below unity (typically $< 10^{-3}$) and flags any silent failure of the Cholesky-fallback chain of Section 6.1.

Score-side diagnostics. The score itself is monitored through:

$$\text{SER} = \frac{\widehat{\text{SE}}_{\text{MC}}}{\widehat{\text{SE}}_{\text{plug-in}}}, \quad \text{ESS}_{\text{abs}}(r) = \frac{(\sum_i |r_i|)^2}{\sum_i r_i^2}, \quad (6.13)$$

$$V_{\text{hat}} = V_{\text{reg}} + V_{\text{corr}} + 2V_{\text{cov}}, \quad \text{Coverage} = \frac{1}{M} \sum_{m=1}^M \mathbf{1}\{|\hat{J}_m - J_\star| \leq 1.96 \widehat{\text{SE}}_m\}. \quad (6.14)$$

Standardised effective ratio. SER compares the dispersion of $\hat{J}$ across Monte-Carlo replications to the average plug-in standard error $\sqrt{\hat{V}/n}$ on the same replications, with $\hat{V}$ derived from (6.14). A value close to unity means the asymptotic distribution is reached at the present sample size; above unity flags a residual bias leaking into the score variance, below unity an over-optimistic plug-in variance (typically from an under-corrected ill-posed Gram).

Effective sample size. ESS_{abs} captures the equivalent number of usable observations under the change of measure induced by the Riesz weights $|r|$; the target is $\text{ESS}_{\text{abs}} \asymp n$ under strong overlap. A sharp drop signals a low-overlap region; the metric therefore reads as an overlap diagnostic, and we use it as such rather than as a tuning criterion.

Variance decomposition. V_{hat} separates the dispersion of the doubly robust score into a plug-in part, an IPW correction part, and their covariance; we read it for the direction and force of the correction. A strongly negative V_{cov} signals that the doubly robust mechanism is working as intended on the dataset under study, whereas a near-zero V_{cov} means the correction adds noise without absorbing any plug-in fluctuation.

Empirical coverage. The fraction of intervals $\hat{J} \pm 1.96 \widehat{\text{SE}}$ that cover the target across the M replications. Under the asymptotic normality of Section 4.5 this fraction should converge to $1 - \alpha = 0.95$; any systematic departure points back to one of the diagnostics above.

Together, these two families provide the language in which we read the simulations of the next section.

6.4 Synthesis

The four methods share the building blocks of Section 6.1 (cross-fit, RFF or Nyström, closed-form policy integration, weight clipping, Cholesky stabilisation), and the bridge-side diagnostics of Section 6.3 provide the empirical analogues of the source index $\mathcal{I}_\beta$. Section 7 tests the methods on three data-generating processes, calibrates their hyperparameters against SER and the bridge-side diagnostics, and reports bias, variance and coverage on $J(\pi_{a,b})$. The qualitative predictions of Section 5 (singularity of the hard contrast, regularity of the stochastic contrast, stability gain from smoothing in A) are read directly off the simulation outputs.

7 Simulations

6

Section 6 fixed the form of the four methods, the finite system each one solves and the metrics that diagnose its numerical stability. We now ask how those methods behave under the operational regimes that Section 5 flagged as fragile. We start with a near-binary process where the stochastic-policy framing is forced by structure, then move to two continuous processes on which the four methods can be compared on equal footing.

7.1 Stochastic policy on a near-binary treatment

The treatment is observed as continuous but is in fact nearly binary: $B \sim \text{Bernoulli}(0.5)$ is perturbed by a narrow Gaussian blur $\varepsilon_U \sim \mathcal{N}(0, 1)$ to give $A = B + \rho_{\text{true}} \varepsilon_U$ with $\rho_{\text{true}} = 0.20$. The response function is the non-linearity $g(t) = \sin(2t)$, the causal target is $J(\pi_{a,\rho_{\text{pol}}})$ for a Gaussian policy of bandwidth ρ_{pol} around the binary value of interest, and the closed form $J_{\text{true}}(\rho_{\text{pol}}) = \exp(-2\rho_{\text{pol}}^2) (\sin 2 - \sin 0)$ provides an analytical reference. We scan ρ_{pol} from the calibrated regime $\rho_{\text{pol}} \approx \rho_{\text{true}}$ towards the singular limit $\rho_{\text{pol}} \rightarrow 0$, with $M = 100$ replications at $n = 1000$ for twelve values between 0.02 and 0.50, and contrast the kernel doubly robust estimator against the Bennett RFF DR estimator.

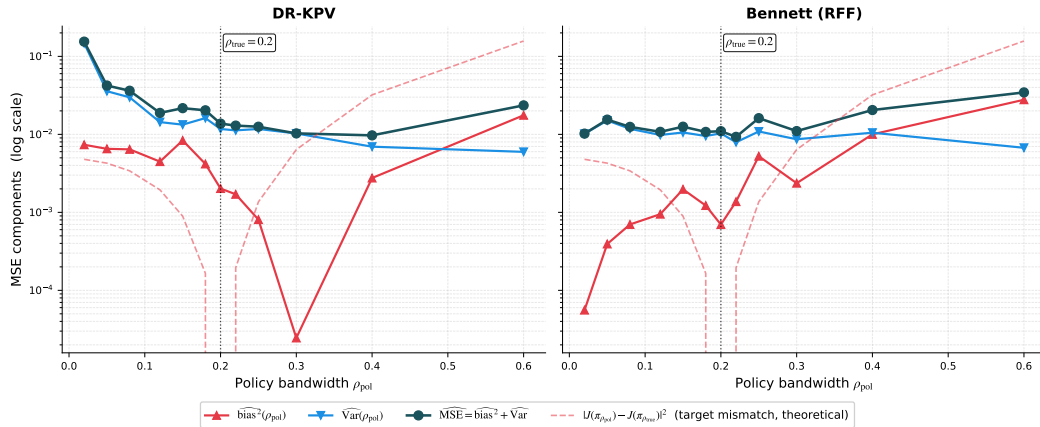


Figure 3: Bias squared, variance and mean squared error of $\hat{J}$ as a function of the policy bandwidth ρ_{pol} , on the coarsened-Gaussian process with $\rho_{\text{true}} = 0.20$, $g(t) = \sin(2t)$, $n = 1000$, $M = 100$. Left, kernel doubly robust estimator. Right, Bennett RFF DR estimator.

The bias reported here is the gap between $\hat{J}$ and the policy-specific target $J_{\text{true}}(\rho_{\text{pol}})$, not the gap to the singular hard contrast. As expected, both methods reach a minimum mean squared error around the calibrated value $\rho_{\text{pol}} \approx \rho_{\text{true}}$. As $\rho_{\text{pol}} \rightarrow 0$ the kernel estimator's variance diverges exactly as Proposition 5.2 predicts: the population IPW weight $K_h(A - a)/f(A | U)$ (estimated by $\hat{q}$) concentrates on the modes $\{0, 1\}$ of the near-binary treatment and the empirical IPW weights are dominated by a small number of

⁶All simulation code, method implementations, and figure-generating pipelines are publicly available at [proximal-causal-inference](https://proximal-causal-inference.github.io).

observations. The Bennett RFF DR estimator remains stable because the integration against the Gaussian policy is in closed form through the spectral damping factor $\exp(-\omega_A^2 \rho_{\text{pol}}^2/2)$ in Equation (6.11). Targeting $J(\pi_{a, \rho_{\text{true}}})$ rather than the hard contrast $\mathbb{E}[Y(1) - Y(0)]$ is therefore the necessary condition for any DR-style estimator on a near-binary treatment, which completes the empirical case for the stochastic-policy framing of Section 2 and Section 5.

7.2 Two continuous processes and the simulation setup

Two processes carry the rest of this section, each translating a distinct causal story. In the Cobb-Douglas log-linear production model, the latent confounder U represents an unobserved quality or productivity shock, the treatment A is a factor of production whose level is contaminated by selection on U , X is an exogenous co-input, and the log-output Y is a linear combination of the three. The Michaelis-Menten dose-response model is non-linear: U stands for an unmeasured physiological sensitivity, the clinician selects the dose A partly on the basis of U so that more sensitive patients receive a different dose, and the response Y is a concave saturating function of the dose at fixed sensitivity. The proxies W and Z are noisy linear measurements of U in both cases.

Cobb-Douglas log-linear (DGP1).

$$\begin{aligned}\log U &\sim \mathcal{N}(0, \sigma_U^2) \\ \log Z &= \log U + \varepsilon_Z \\ \log W &= \log U + \varepsilon_W \\ \log A &= \gamma_K \log U + \eta_A \\ \log X &\sim \mathcal{N}(\mu_L, \sigma_L^2) \\ \log Y &= \log U + \alpha \log A + (1 - \alpha) \log X + \varepsilon_Y\end{aligned}$$

The structural elasticity $\alpha = 0.3$ plays the role of the average treatment effect; both bridges admit a closed form.⁷

The signal-to-noise ratios are set by the proxy noise variances, $\text{SNR}_W = \sigma_U^2 / (\sigma_U^2 + \sigma_{\varepsilon_W}^2)$ and symmetrically for Z ; the target dose grid is $a \in \{1.8, 2.2, 2.6, 3.0\}$ on DGP2 (reference dose $a = 2.6$ at the empirical mode of A), and a point-mass policy at $a = 0$ on DGP1. The reference cell of DGP2 is $n = 2000$, $\text{SNR}_W = \text{SNR}_Z = 0.95$, $a = 2.6$; all metrics of Section 6.3 are computed on $M \geq 100$ disjoint Monte-Carlo seeds with full refit (cross-fitting included) at each replication; this discipline on M is motivated by past small- M winner failures documented in Section 7.9.

Michaelis-Menten dose-response (DGP2).

$$\begin{aligned}U &\sim \text{LogNormal}(0, \sigma_U^2) \\ Z &= U + \varepsilon_Z \\ W &= U + \varepsilon_W \\ A &= a_0 + sU + \eta_A \\ Y &= \frac{V_{\max} U A}{K_m U + A} + \varepsilon_Y\end{aligned}$$

The dose response is strictly concave in A for fixed $U > 0$; the treatment bridge admits a closed form, the outcome bridge does not.⁸

7.3 Convergence and scale

Two prerequisites condition every comparison that follows: the Monte-Carlo budget $M = 200$ at the reference cell must shrink the bootstrap intervals on the metrics to a negligible width, and the sample size $n = 2000$ must reach the asymptotic regime of the methods.⁹

⁷For DGP1, the parameters are $\alpha = 0.3$, $\gamma_K = 1$, $\sigma_K^2 = \text{Var}(\eta_A) = 1$, $\sigma_Z^2 = \text{Var}(\varepsilon_Z)$. The bridges are $h_0(W, A, X) = W + \alpha A + (1 - \alpha)X$ and $q_0(Z, A) = \sqrt{2\pi\sigma_K^4/D_Z^{(1)}} \exp[(A - \gamma_K Z)^2 / (2D_Z^{(1)})]$ with $D_Z^{(1)} = \sigma_K^2 + \gamma_K^2 \sigma_Z^2$.

⁸For DGP2, the parameters are $V_{\max} = 2$, $K_m = 1$, $a_0 = 2$, $s = 0.3$, $\sigma_{\eta_A} = 0.6$, $\sigma_U = 1$. The treatment bridge is $q_0(Z, A) = \sqrt{2\pi\sigma_{\eta_A}^4/D_Z^{(2)}} \exp[(A - a_0 - sZ)^2 / (2D_Z^{(2)})]$ with $D_Z^{(2)} = \sigma_{\eta_A}^2 + s^2 \sigma_Z^2$.

⁹The top panel of Figure 4 is produced by drawing 100 bootstrap sub-samples of size M from a pool of 200 replications and recomputing bse_{ref} and the empirical coverage on each sub-sample; the band reports the 90% interval across these bootstraps. The broader DGP2 grid uses $M = 100$ in production and the proxy-quality grid of Figure 9 uses $M = 80$.

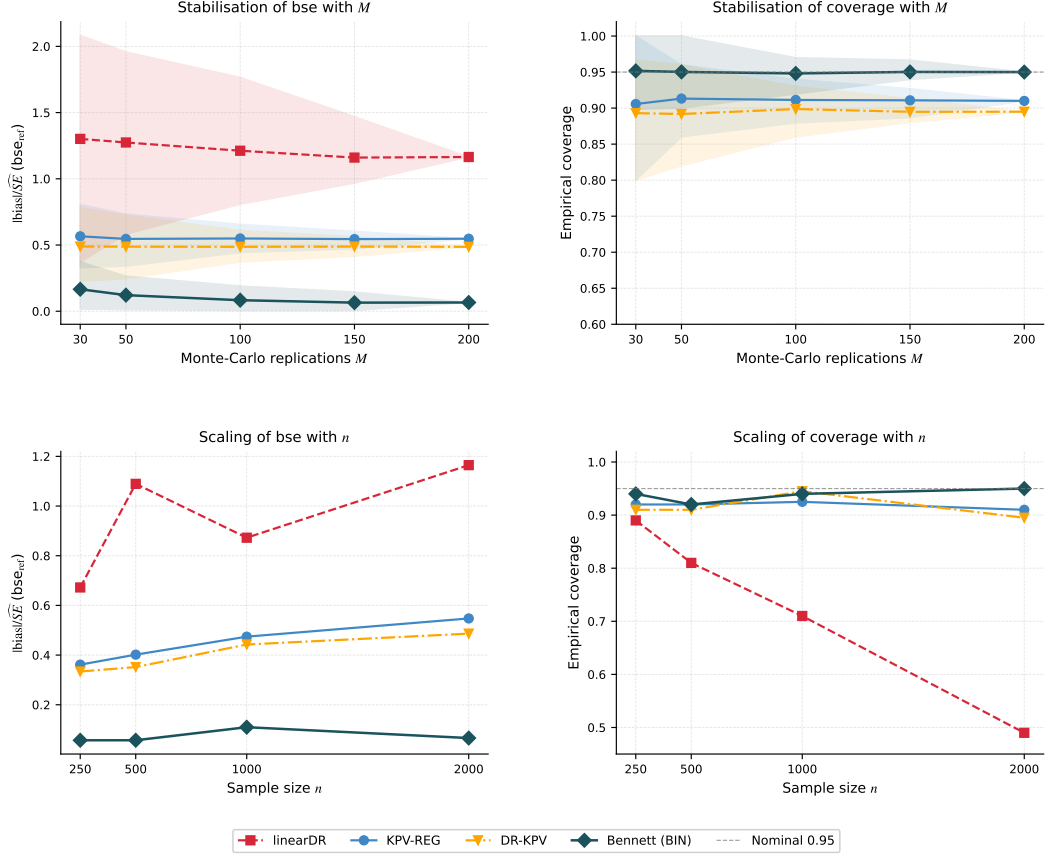


Figure 4: Convergence and scale on the Michaelis-Menten process at the reference cell $\text{SNR}_W = \text{SNR}_Z = 0.95$, $a = 2.6$. Top, stabilisation of bse_{ref} and empirical coverage with the number of Monte-Carlo replications M (bootstrap 90% band over 100 sub-samples). Bottom, scaling of the same metrics with the sample size n at $M = 100$, one curve per method.

By $M = 100$ the bootstrap interval on coverage tightens to about ± 0.05 and is negligible by $M = 200$, which validates $M \in \{100, 200\}$ for the rest of this section. Beyond $n = 1000$ the curves stabilise for the methods that are correctly specified, and at $n = 2000$ their empirical coverage approaches the nominal 0.95. The convergence is not universal: the linear two-stage estimator does not reach nominal coverage even at large n because its parametric bridge is misspecified against the concave non-linearity, and on degraded-proxy cells the linear DR drops below 0.85 at $\text{SNR} \leq 0.70$ while Bennett RFF DR drops at $\text{SNR} < 0.85$. The reference cell is therefore representative of an operational regime only for correctly specified methods at strong signal-to-noise.

7.4 Double robustness in practice

We probe the doubly robust correction in two complementary movements: a degraded-bridge experiment that reads off the direction and intensity of the correction, and a regularisation stress test that varies λ_h and the second-bridge regulariser independently to ask whether the functional remains identifiable when either bridge drifts.

To make the correction work, we pick the most difficult proxy regime of the Michaelis-Menten process, $\text{SNR}_W = \text{SNR}_Z = 0.50$ at $n = 1000$, and condition the bias estimate on the size of the regression error. Define the threshold-filtered seed set $\mathcal{M}_\tau = \{i : |\hat{J}_{\text{REG}}^{(i)} - J_{\text{true}}| > \tau \widehat{\text{SE}}^{(i)}\}$, and let $\hat{b}_{\text{REG}}(\tau), \hat{b}_{\text{DR}}(\tau)$ denote the cross-seed biases restricted to $\mathcal{M}_\tau$. The conditional BiasFix

$$\widehat{\text{BiasFix}}(\tau) = \frac{|\hat{b}_{\text{REG}}(\tau)| - |\hat{b}_{\text{DR}}(\tau)|}{|\hat{b}_{\text{REG}}(\tau)|} \quad (7.1)$$

isolates the structural component of the bias from the Monte-Carlo noise: a positive value means the DR correction reduces the structural bias, a negative one means it amplifies it, zero means it is neutral.

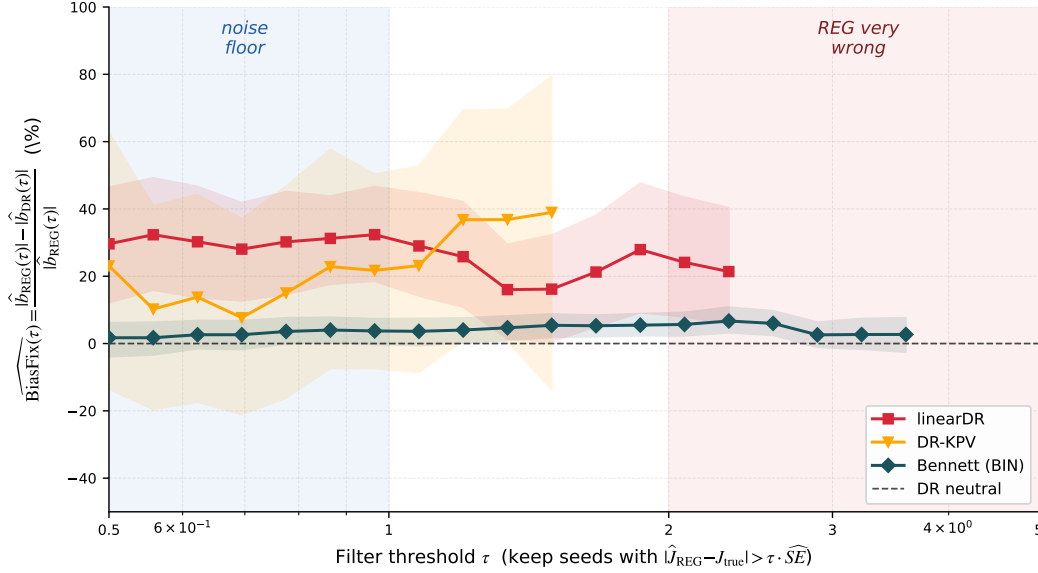


Figure 5: Conditional BiasFix $\widehat{\text{BiasFix}}(\tau)$ as a function of the filtering threshold τ (log scale), for the three doubly robust estimators on the most degraded proxy cell of the Michaelis-Menten process ($\text{SNR}_W = \text{SNR}_Z = 0.50$, $n = 1000$, $a = 2.6$). Bands are bootstrap 95% intervals; the dashed line at zero marks a neutral correction.

The linear DR estimator shows a stable BiasFix between +25% and +35% across all τ , since its parametric bridge is so severely misspecified against the Michaelis-Menten concavity that the correction always finds a structural bias to fix. The kernel DR estimator climbs from about +10% at $\tau = 0.5$ to about +40% at $\tau = 2$, the correction is positive and grows with the size of the regression error, exactly as the cross-product structure of the bias predicts. The Bennett RFF DR estimator hovers between 0 and +5%, since even on this degraded cell its bridge is so close to the truth that the correction has almost nothing to fix. The ranking linear > kernel > Bennett RFF follows the ordering of the regression bridge fragility.¹⁰

The second movement varies λ_h and the second-bridge regulariser (λ_q or λ_r) independently across three orders of magnitude on a 3×3 grid, scoring each cell by

$$\text{score}(\lambda_h, \lambda_q \text{ or } r) = \text{bse}_{\text{ref}} + |\text{cov} - 0.95| \quad (7.2)$$

at the reference cell of the Michaelis-Menten process.

¹⁰The decomposition $\text{MSE}_{\text{DR}} - \text{MSE}_{\text{REG}} = (b_{\text{DR}}^2 - b_{\text{REG}}^2) + (\sigma_{\text{DR}}^2 - \sigma_{\text{REG}}^2)$ of Section 4 reminds us that the correction reduces the bias but also adds variance: the IPW weight $\hat{q}$ injects empirical noise. The DR estimator is therefore better in mean squared error only when the reduction of structural bias dominates the induced variance inflation.

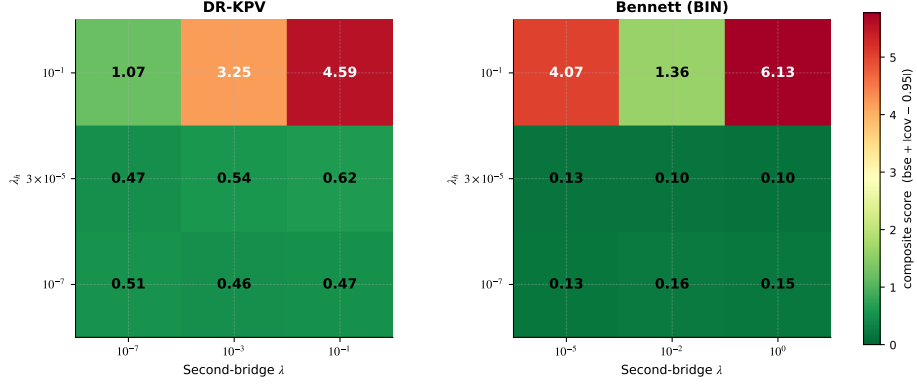


Figure 6: Composite score (7.2) over a 3×3 grid of regularisations (λ_h, λ_q) (left, kernel DR) and (λ_h, λ_r) (right, Bennett RFF DR). Reference cell of the Michaelis-Menten process, $n = 2000$, $\text{SNR}_W = \text{SNR}_Z = 0.95$, $a = 2.6$.

The pattern is asymmetric: the lower two rows of both heatmaps remain green for any λ_q or λ_r , while the top row at $\lambda_h = 10^{-1}$ collapses with composite scores between 1.07 and 6.13. Double robustness therefore requires a minimal quality floor on the outcome bridge but lets the second bridge drift across orders of magnitude, in line with the cross-product bias of Section 4.

7.5 Bridge–functional decoupling on the linear process

A short calibration check on the Cobb-Douglas process anchors the pipeline before turning to a more revealing question: all proximal-aware methods (oracle, linear DR, kernel DR, Bennett RFF DR) perform well with $\text{bse} < 0.1$ and coverages above 0.90, while the naive regression that omits W reaches $\text{bse} \approx 0.87$ with coverage about 0.36. No reasonable method breaks on a correctly specified linear setting. The interesting object is then the relation between the error of $\hat{h}$ on the bridge in L^2 and the error of $\hat{J}$ on the functional. For six methods (oracle, naive without W , linear regression, linear DR, kernel DR, Bennett RFF DR) and $M = 30$ replications at $n = 1000$, we extract $\hat{h}$, evaluate $\|\hat{h} - h_0\|_{L^2(P)}$ on a held-out test set of size $N = 10^5$, and pair it with the absolute functional error $|\hat{J}(\pi_{a=0}) - J_{\text{true}}(\pi_{a=0})|$.

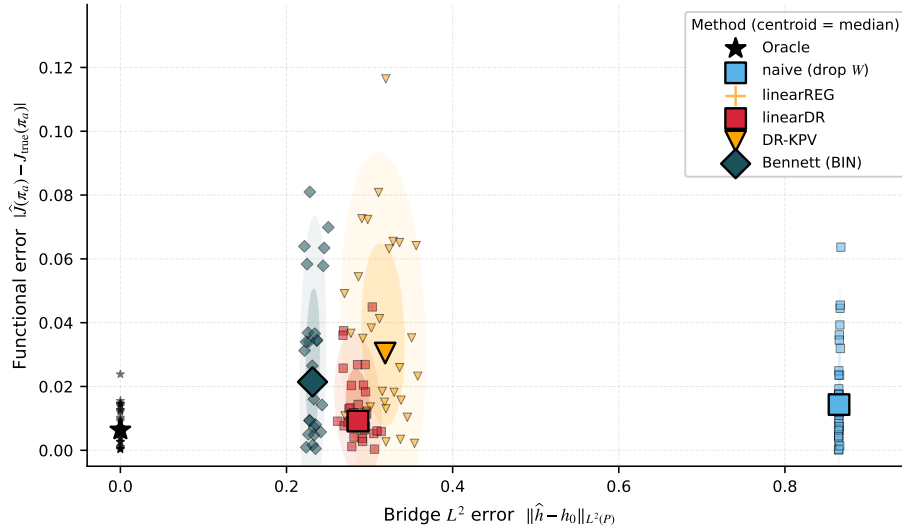


Figure 7: Bridge L^2 error versus functional error per method on the Cobb-Douglas process at $n = 1000$, $M = 30$, $a = 0$. Translucent markers are individual seeds, solid markers are method medians, ellipses are 1σ and 2σ envelopes of the empirical seed-by-seed covariance.

Somewhat surprisingly, the naive regression has an L^2 bridge error three to four times larger than every other method (≈ 0.87 against 0.2 to 0.3), yet its functional error is almost identical to the others (≈ 0.02). The naive bridge is markedly far from h_0 across the joint domain (W, A, X) , but this gap absorbs into the

integration $\int \hat{h} \pi_a$ and does not propagate to the functional. On this process J is therefore more identifiable than the bridge, which justifies the functional-targeting approach (Bennett RFF DR, kernel DR) over the bridge-fitting approach as a default methodological stance.¹¹

7.6 Estimand distributions and proxy sensitivity

Two figures probe the same operational question at different grains. The dose comparison contrasts a central dose $a = 2.6$, where the empirical density of A is high, with a boundary dose $a = 1.8$ sitting in a low-density region that forces the estimator to extrapolate; the proxy-quality map then expands the comparison over the full 4×4 grid of $(\text{SNR}_W, \text{SNR}_Z)$ combinations.

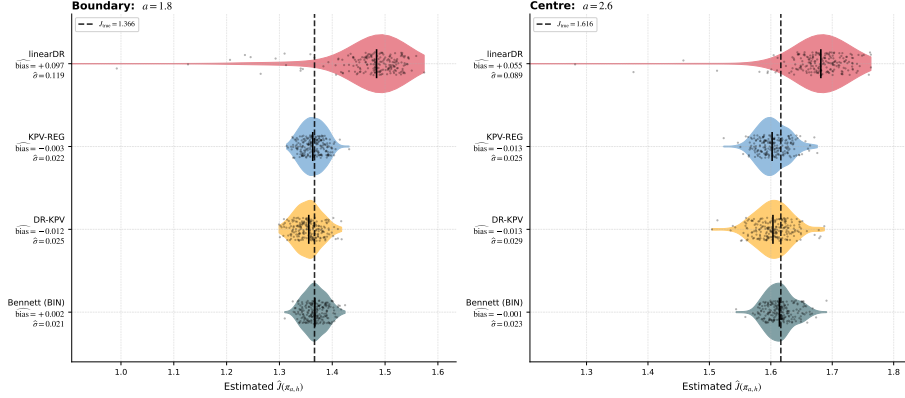


Figure 8: Empirical distributions of $\hat{J}(\pi_{a,b})$ on $M = 200$ replications, four methods (linear DR, KPV-REG, kernel DR, Bennett RFF DR), at the boundary dose $a = 1.8$ (left) and the central dose $a = 2.6$ (right) of the Michaelis-Menten process, $n = 1000$, $\text{SNR}_W = \text{SNR}_Z = 0.95$. Annotations report the empirical bias $\hat{b}$ and seed-by-seed standard deviation $\hat{\sigma}$; the dashed vertical line marks $J_{\text{true}}(\pi_{a,b})$. At the central dose the proximal-aware methods are tight ($\hat{\sigma} \in [0.023, 0.029]$) while the linear DR is shifted by $\hat{b} = +0.055$ at $\hat{\sigma} = 0.089$. At the boundary the linear DR explodes to $\hat{b} = +0.097$, $\hat{\sigma} = 0.119$ while proximal-aware methods stay within $\hat{\sigma} \leq 0.030$.

The boundary dose amplifies dispersion for every method, but it exposes the parametric misspecification of the linear DR without breaking the proximal-aware methods. Inference quality tracks the overlap between target dose and the empirical support of A , a point we will see again on real data.

¹¹The decoupling is specific to proxy-omission misspecification (bridge error can be harmful in other settings, just not here). Coverage stays orthogonal anyway: 0.36 for the naive method, above 0.90 for the others.

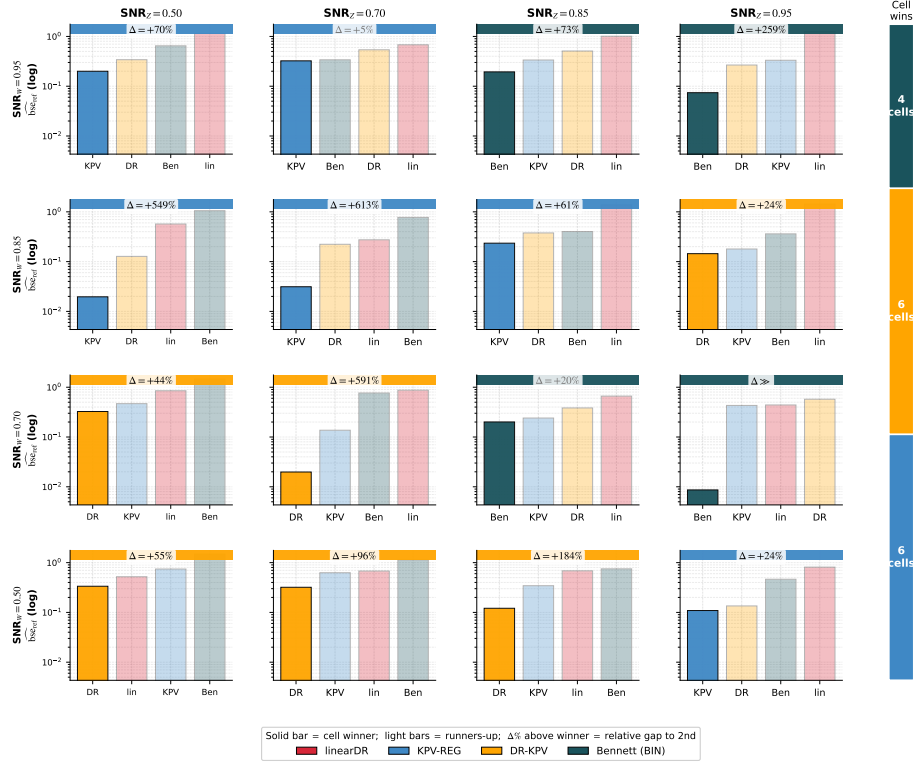


Figure 9: Per-cell ranking of four methods by bse_{ref} on the proxy-quality grid $(SNR_W, SNR_Z) \in \{0.50, 0.70, 0.85, 0.95\}^2$. Each cell holds four bars sorted by ascending bse on log scale; the coloured strip on top encodes the winning method, the relative gap $\Delta\%$ to the runner-up is annotated above. The right-hand strip stacks the cells won by each method. Michaelis-Menten process, $n = 1000$, $M = 80$, $a = 2.6$; hyperparameters frozen at the central calibration cell.

Each method is evaluated with the hyperparameters that won at the central cell $(SNR_W, SNR_Z) = (0.95, 0.95)$ without re-tuning per cell, so the figure measures robustness to hyperparameter transfer rather than maximal capability.

The Bennett RFF DR estimator dominates the four high-signal-to-noise cells of the upper-right quadrant with relative gaps up to +259% at the reference cell, which is its zone of optimal tuning rather than its zone of robustness. The kernel DR estimator picks up around five intermediate and low-signal-to-noise cells where neither the regression plug-in nor the Bennett RFF DR estimator is competitive. Its kernel regularisation auto-adapts to the empirical Gram matrices, whereas the random-feature method carries fixed hyperparameters (m_h, ℓ_h, λ_h) that do not transfer. Around five further cells go to the kernel regression plug-in in the regime where W is strong but Z is degraded: the IPW correction through $\hat{q}$ penalises the doubly robust estimator (the q -bridge is built from Z) while a regression plug-in on W remains informative as long as W is strong. As soon as the Michaelis-Menten non-linearity is active the linear two-stage estimator plateaus and loses everywhere. At equal signal-to-noise, Z is marginally more binding than W for the Bennett RFF DR estimator, consistent with the duality between the efficient influence function and the Riesz representer of Section 4; the Pareto pattern is solid, individual cell values are accurate to about ± 0.05 .

We read the grid as a regime map. No single method wins across all cells. A practitioner who does not know the regime should default to the kernel DR estimator, whose performance is less peaked than the Bennett RFF DR estimator at the centre but covers a broader zone of proxy qualities and sample sizes; the Bennett RFF DR estimator becomes the recommended choice only when both SNR_W and SNR_Z are independently known to be strong.

7.7 Spectral diagnostics for blind tuning

We test which in-sample diagnostics introduced in Section 6.3 actually predict the out-of-sample inference quality by computing Spearman correlations between candidate diagnostics and the gold-standard bse_{ref} across 22 simulation cells on the Michaelis-Menten process.

Bennett RFF DR			Kernel DR (DRKPV)		
Diagnostic	$\rho(\text{bse})$	p	Diagnostic	$\rho(\text{bse})$	p
$r_{\text{eff},h}$	-0.83	< 0.001	κ_r	-0.94	0.005
$r_{\text{eff},r}$	-0.81	< 0.001	SER	-0.49	0.022
κ_r	-0.81	< 0.001	RR_{ref}	-0.46	0.031
res_h	+0.76	< 0.001	res_r	-0.41	0.060
κ_h	-0.64	0.001			
res_r	+0.62	0.002			
SER	-0.52	0.013			

Table 1: Spearman correlations between in-sample diagnostics and the bias-to-SE ratio bse_{ref} across 22 cells of the Michaelis-Menten process. Bennett RFF DR (left) admits multiple actionable diagnostics with $|\rho| > 0.6$ at $p < 0.005$; Kernel DR (right) is dominated by κ_r at $|\rho| = 0.94$. The ESS_{min} -ratio is omitted from both lists because its correlation with bse is positive (Bennett: +0.84; DRKPV: +0.41): a more localised policy reduces it while inflating the structural bias, so the diagnostic measures policy concentration rather than bridge quality and cannot serve as a tuning criterion.

For Bennett RFF DR three diagnostics reach $|\rho| > 0.8$: the effective ranks of h and of the Riesz (when the bridge is ill-conditioned more s_j fall below λ , pulling r_{eff} down) and the conditioning of the Riesz Gram. For Kernel DR the spectral fields specific to RFF are not defined, but κ_r alone predicts bse_{ref} at $|\rho| = 0.94$. Section 7.8 benchmarks the resulting protocol against an oracle.

7.8 Blind versus oracle hyperparameter calibration

The blind tuning protocol uses only in-sample diagnostics and the SER, never the unknown J_{true} . Five stages structure the search: 1D coarse scans on each hyperparameter axis, 2D refinement on a product grid, cross-cell consistency across three signal-to-noise levels, disjoint-seed validation gated by $\text{SER} \in [0.85, 1.15]$ at the reference cell, and a regime-robustness check on four stress cells.

Kernel DR protocol. The blind tuner sweeps $(\lambda_h, \ell_h, \lambda_Q)$, with a coarse-stage gate $\kappa_h < 10^8$ and a Riesz residual cap $< 10^{-1}$. The winner $(\lambda_h, \ell_h, \lambda_Q) = (10^{-5}, 5, 10^{-5})$ passes the SER gate at all four Stage 5 stress cells; we call this a *robust envelope*.

Bennett RFF DR protocol. On $(\lambda_h, \ell_h, \gamma_{\text{critic}}, \lambda_r)$ at fixed $m_h = 2000, m_c = 500$, an additional *interpolation gate* $\lambda_h \times m_h > 10^{-2}$ hard-fails any candidate whose feature dimension is too large for the chosen ridge to keep the Cholesky away from singularity. The winner $(\lambda_h, \ell_h, \gamma_{\text{critic}}, \lambda_r) = (10^{-5}, 1.5, 10^{-5}, 10^{-3})$ passes the SER gate only at the reference cell, a *regime-conditional envelope*.¹²

¹²The two ℓ_h scales are not directly comparable: Kernel DR's ℓ_h is an RBF length-scale on raw covariates, whereas Bennett's ℓ_h enters the Fourier frequency variance of the RFF feature sampler.

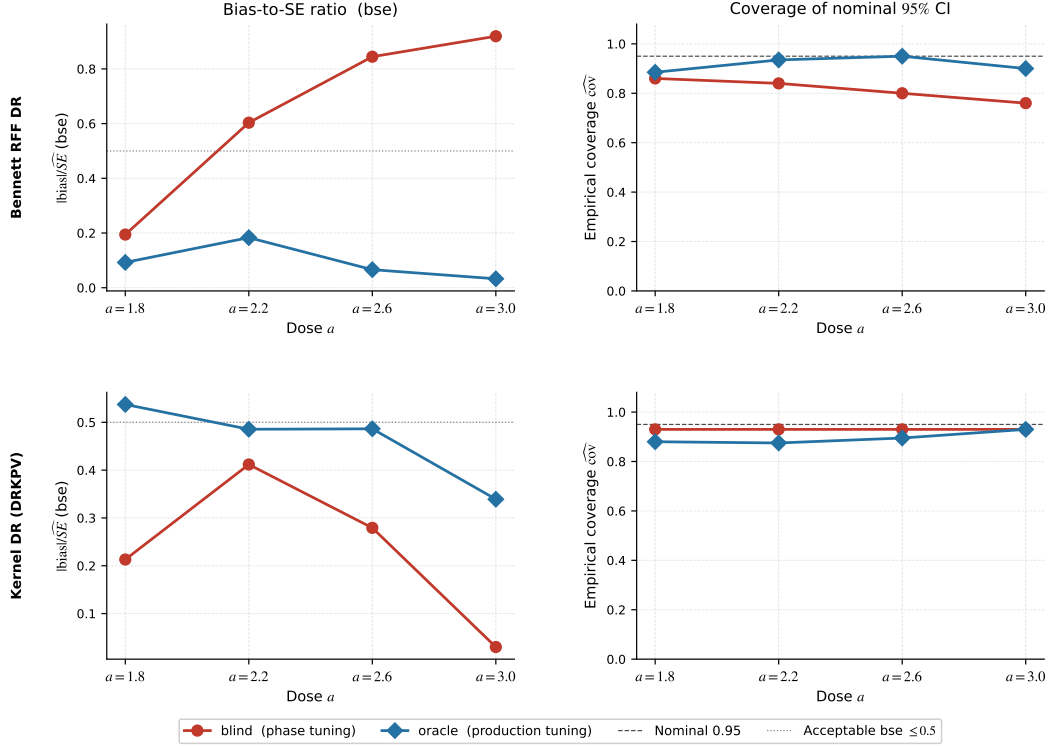


Figure 10: Blind versus oracle calibration on the Michaelis-Menten process at the reference cell ($n = 2000$, $\text{SNR}_W = \text{SNR}_Z = 0.95$, $M = 100$). Top row: Bennett RFF DR estimator. Bottom row: kernel DR estimator. Left column: bse_{ref} per dose. Right column: empirical coverage per dose, with the nominal 0.95 line.

For the Bennett RFF DR estimator the blind protocol matches the oracle at the central dose $a = 2.6$ but its performance degrades sharply away from it. At the intermediate dose $a = 2.2$ the blind bse_{ref} rises to about 0.60 against 0.12 for the oracle; at the boundary dose $a = 3.0$ blind bse_{ref} reaches about 0.92 against 0.04 for the oracle, with coverage dropping to 0.74. The residual gap concentrates on the lengthscale ℓ_h , which no in-sample diagnostic discriminates finely: the role of ℓ_h manifests in the projection of the bridge onto the support of A , which is intrinsically a question of out-of-sample extrapolation. The regularisation λ_h is by contrast identifiable blindly through the interpolation gate, so it is retained correctly.

For the kernel DR estimator the blind protocol is conservative by construction. Coverage stays close to nominal across the dose grid, and the gap to the oracle is tight in absolute terms.

As for regularisation parameters λ , they are identifiable blindly: a structural gate on $\lambda \times \dim(\text{feature})$ excludes the pathological zones and the in-sample diagnostics discriminate the remaining candidates. Lengthscales ℓ are not. No in-sample diagnostic ranks ℓ -candidates reliably, and the role of ℓ in out-of-sample extrapolation is structurally invisible to any in-sample score. On a real dataset where J_{true} is unavailable, the lengthscale must be anchored by a resource external to the blind tuner: a held-out subsample, domain knowledge, or a structured sensitivity analysis. This λ/ℓ asymmetry is what the next section examines on real data.

7.9 Limitations

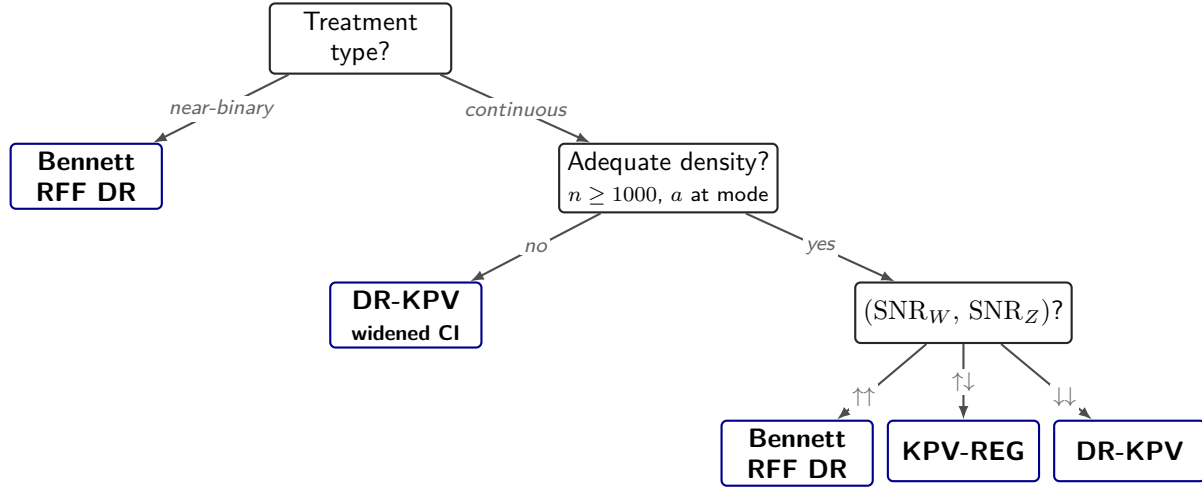
We adopted the discipline of $M \geq 100$ disjoint seeds and an SER gate at validation time in response to seven documented episodes during the design of the campaign where a configuration apparently optimal at small M failed to survive an independent revalidation; none of the conclusions reported above involves a small- M winner. The lengthscale ℓ_h is not identifiable through any in-sample diagnostic we tested, which limits the transfer of our protocol to settings where an external anchor for ℓ_h is available, a caveat that is the principal connection with Section 8. The Bennett RFF DR estimator becomes unreliable at $\text{SNR} < 0.85$ on the Michaelis-Menten process, with bse_{ref} above 0.7 at $\text{SNR} = 0.80$; the kernel DR estimator remains the

safe fallback in that regime. Lastly, we ran the proxy-quality grid at $M = 80$ per cell rather than $M = 200$, which was already four days of compute. The Pareto pattern across the 16 cells is solid, but individual cell values are accurate to about ± 0.05 . That is enough to anchor the regime map without overinterpreting the precise relative gaps cell by cell.

7.10 Practitioner protocol

We translate the empirical regularities documented above into a compact decision protocol. Figure 11 lays out the method choice along three branching questions; Table 2 lists the diagnostic gates that every retained configuration must pass at validation time before deployment.

Prerequisite: $M \geq 100$ disjoint seeds, full cross-fitted refit per replication, hyperparameters anchored externally.



Lengthscale ℓ_h is not identifiable from in-sample diagnostics. Anchor externally (hold-out, domain knowledge, sensitivity analysis) before deployment.

Figure 11: Practitioner decision protocol synthesising the empirical findings of Section 7.4, Section 7.6, and Section 7.8. Branch labels $\uparrow\uparrow$, $\uparrow\downarrow$, $\downarrow\downarrow$ encode high (both ≥ 0.85), asymmetric (W strong, Z weak), and low (both ≤ 0.70) signal-to-noise regimes on the proxies. Method labels: Bennett random-feature doubly robust (Bennett RFF DR), kernel doubly robust (DR-KPV), kernel plug-in regression (KPV-REG).

Gate	Threshold	If fail
SER at validation cell	$\in [0.85, 1.15]$	Downgrade to DR-KPV or expand n
κ_h	$< 10^8$	Re-tune λ_h
res_h	$< 10^{-1}$	Strengthen regularisation
$\lambda_h \times \dim(\Phi_H)$	$> 10^{-2}$	Excludes interpolation collapse

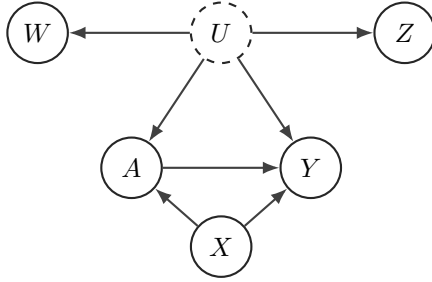
Table 2: Pre-deployment diagnostic gates. The diagnostics are defined in Section 6.3; their empirical predictive value is established in Table 1. Any gate failure triggers the action listed and a return to the decision tree of Figure 11.

8 Applications

Section 7 showed that the blind tuning protocol holds in simulation. How does it behave on real data, where the data-generating process is unknown and the proxies are not constructed by design? We answer on two datasets drawn from very different causal settings, on which the classical literature provides external anchors to triangulate the proximal estimate against.

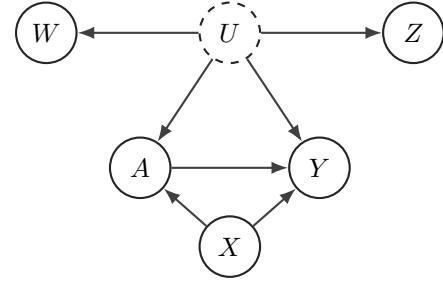
8.1 Two applications and their causal scaffolds

The NLSY79 cohort answers the canonical labour-economics question of the return to schooling, with father's years of completed schooling as an outcome-inducing proxy and a lagged household record as a treatment-inducing proxy. Connors Jr. et al. [30]'s right-heart-catheterisation (RHC) study is the second application: we estimate the effect of inserting a pulmonary artery catheter on 30-day mortality in critical-care patients, with admission pH as W and the $\text{PaO}_2/\text{FiO}_2$ ratio as Z , both pre-procedure markers of disease severity. The literature anchors are the IV/twins consensus $[+8\%, +12\%]$ for NLSY79 [31, 32] and the randomised-controlled-trial consensus of no clear mortality effect for RHC, summarised by the Cochrane band $[-0.03, +0.02]$ on the probability difference scale [33, 34, 35]. Sample sizes after listwise deletion are $N = 4977$ on NLSY79 and $N = 5735$ on RHC. We set our success criterion as follows: the blind-tuning winner yields an estimate inside or compatible with the literature reference range, under blind diagnostics that pass internal validation.



(a) NLSY79 return to schooling.

U : ability and family human capital;
 W : father's HGC; Z : lagged household HGC;
 A : own HGC; Y : log hourly wage;
 X : race, sex.



(b) RHC critical-care mortality.

U : latent disease severity;
 W : admission serum pH; Z : $\text{PaO}_2/\text{FiO}_2$;
 A : RHC indicator; Y : 30-day mortality;
 X : age, mean blood pressure.

Figure 12: Proximal causal scaffolds intended for the two applications, in the formalism of Section 2. The unobserved confounder U is dashed; the proxies W (NCO) and Z (NCE) play the structural roles encoded in Assumption 2.4 and Assumption 2.5 respectively, and the absent edges express the corresponding exclusions. The father design on NLSY79 partially violates the outcome-proxy exclusion of Assumption 2.4 through multi-channel paths from W to Y , a point we revisit in Section 8.4.

8.2 Methodology adaptations

We modify the simulation pipeline at four points. The data-generating process is replaced by a nonparametric bootstrap with replacement of size N , repeated $M = 100$ times, to produce pseudo-replications in the absence of a known sampling distribution; the sample-error ratio SER is uninformative on bootstrap data and is dropped from the blind score, which is SER-free for the Bennett RFF DR estimator by construction.

The substantively natural treatment of the observable covariates X would have been to absorb them non-linearly into the bridge feature map, on the same footing as the proxies. This adds substantial complications to the tuning protocol that we did not pursue here. We tested instead a simpler linear-residualisation step (Frisch-Waugh-Lovell partial-out via ridge regression), and on RHC it collapsed the proxy structure: the bridge fit residual $\|A_h\hat{\beta} - d_h\|/\|d_h\|$ jumped from $\sim 5 \times 10^{-3}$ to ~ 0.5 . Whether a fuller kernelisation of X would behave similarly is left open. We proceed without residualisation on both datasets.

The kernel doubly robust estimator is excluded for computational reasons. Its q -bridge condition-number diagnostic scales as $O(N^3)$ per fold per replication, several days on either dataset for a full Stage 1 axis sweep at $M = 100$, while the Bennett RFF DR estimator runs in $O(N)$ via a fixed-size random-feature Gram. We thus benchmark the linear two-stage proximal estimator, KPV-REG, and Bennett RFF DR on both datasets, with OLS and OLS-on-covariates baselines added on RHC for direct comparison with the Connors literature.

We finally layer an adaptive checkpoint structure on top of the five-stage protocol of Section 7.8: three inspection points (pre-tuning, post-validation, post-stress) at which the analyst decides GO, CAUTION, or ABORT before committing further compute. The pre-tuning inspection caught a third candidate dataset whose proxy- U product correlation was near zero, suggesting identification was unlikely to hold. A boundary-

aware refinement, called Win A, re-runs the blind sub-protocol on a widened axis whenever a Stage 4 winner sits at a grid edge, with an explicit acceptance criterion that the new winner improve the blind score by more than $\epsilon = 0.01$. This makes any boundary-driven adjustment transparent rather than discretionary.

The procedure is identical on both datasets (Stage 1 axis scans with structural gates, Stages 2–5 refinement and disjoint-seed validation); one of the two datasets triggered Win A, the other did not.

8.3 Triangulation and dose-response figures

Figure 13 positions each method relative to the literature reference; Figure 14 traces the Bennett RFF DR dose-response curves with bootstrap envelopes. Section 8.4 reads them dataset by dataset.

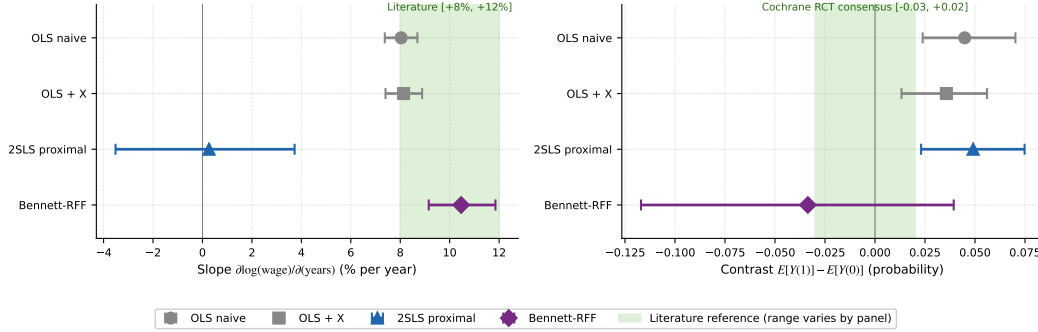


Figure 13: Triangulation of the slope estimate on the two applications. Methods plotted: linear two-stage proximal, KPV-REG, Bennett RFF DR (and OLS, OLS-on-covariates on RHC). Vertical green bands: literature reference ranges (NLSY79 IV/twins consensus $[+8\%, +12\%]$ from [31, 32]; RHC randomised-trial consensus $[-0.03, +0.02]$ on the probability-difference scale from [33, 34, 35]). Bars: 95% bootstrap CIs ($M = 100$).

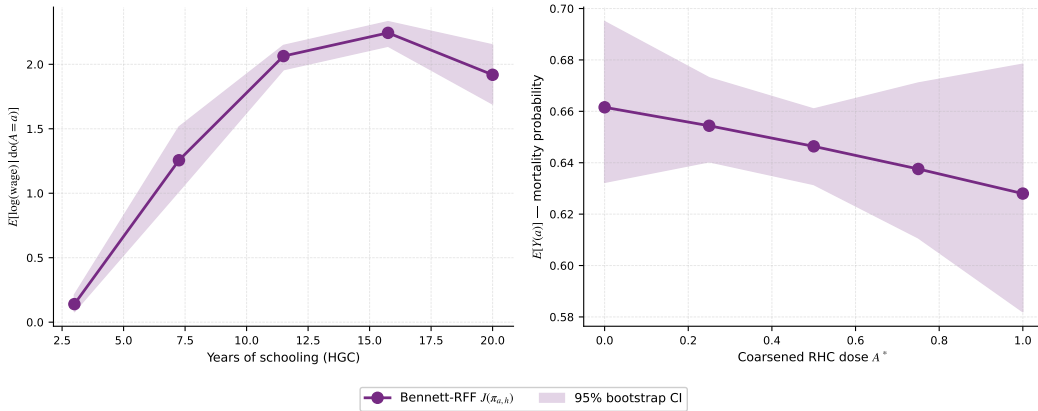


Figure 14: Bennett RFF DR dose-response curves $J(\pi_{a,b})$ on the two applications, with 95% bootstrap envelopes ($M = 100$ replications). NLSY79: monotone increasing across the schooling range, with the envelope tightest near the median dose. RHC: gently decreasing in the coarsened-dose direction a^* , with widening envelope near the treated arm.

8.4 What the figures reveal on each dataset

We observe the same pattern on both datasets: the linear estimators fail to recover the literature consensus, and the Bennett RFF DR estimator is the only one compatible with it (inside the band on NLSY79; with a confidence interval that envelopes the band on RHC).

On NLSY79 the linear two-stage proximal estimator returns $+0.3\%$ per year of schooling with a CI $[-3.5, +3.7]$ that includes zero, well outside the IV/twins consensus $[+8\%, +12\%]$. The Bennett RFF DR estimator with $m_h = 2000$ random Fourier features captures the local curvature of the schooling-wage relationship and recovers $+10.4\%$ with CI $[+9.2, +11.9]$, fully inside the consensus and closer to Card [31] and Ashenfelter and Krueger [32] than to the canonical Mincer $+8\%$ [36]. The pattern is consistent with the reading that the non-linear bridge captures variation the linear baselines miss, although other interpretations

remain compatible. The Win A widened-grid refinement was triggered on this dataset because the Stage 4 winner sat at the lower edge of the lengthscale grid; the widened sub-protocol selected $\ell_h = 0.5$ with a score improvement of 0.039 on disjoint validation seeds, above the acceptance threshold, and the slope quoted above is the resulting blind-score winner.

On RHC the linear estimators all point in the wrong direction. OLS gives +4.4% excess mortality (CI [+2.4, +7.0]%, the result that drove the original Connors controversy), conditioning on age and mean blood pressure barely moves the estimate to +3.5% (CI [+1.3, +5.6]%), and the linear two-stage proximal estimator stays at +4.9% (CI [+2.3, +7.5]%). All three are several standard errors above the Cochrane band of no clear mortality effect, the binding linear feature being that RHC is a binary contrast on which a linear bridge cannot represent the heterogeneity that drives the non-causal correlation. The Bennett RFF DR estimator returns -3.4% with wide CI [-11.7, +3.9]%, which fully encloses the RCT consensus and is sign-compatible with the proximal-causal estimate of Tchetgen Tchetgen et al. [7]. The dose-response curve is gently decreasing across the coarsened dose direction with envelope widest near the treated arm, consistent with a small protective or null effect rather than the sharp positive effect that the linear models produce.

The linear baselines fail in opposite ways: underestimation on NLSY79 (the slope nearly vanishes once the linear bridge partials out the proxy) and overestimation on RHC (the linear bridge amplifies the OLS confounding rather than correcting it). The cost was substantial. The NLSY79 father design partially violates the outcome-proxy exclusion Assumption 2.4, since father's schooling reaches the individual's wage through multiple channels rather than only via the latent U , and no alternative W or Z is available in the dataset. On RHC, we used alternative proxies W_2 (hematocrit) and Z_2 (PaCO₂) following Cui et al. [8] at Stage 5, and the substitution sensitivity remained below 2% of the central estimate. The lengthscale ℓ_h remained the residual non-identifiable parameter, as Section 7.8 predicted. We read these two applications as preliminary evidence that PCI is operationally usable on post-hoc data when the proxies, however imperfect, can be defended substantively. The limits we encountered make the case for designing future studies with proximal identification in mind.

8.5 PCI in the causal-inference landscape

What we want to answer here is whether PCI is realistic for the problem at hand. Backdoor adjustment and its modern doubly robust extensions [24, 37] are the operational standard when observable covariates plausibly capture the confounding structure; instrumental variables and quasi-experimental designs work when a credible exogenous variation is available; both are silent when serious unmeasured confounding remains and neither an instrument nor a clean design exploit is available.

PCI's distinctive ingredient is a pair of negative controls satisfying conditional independence on the latent U rather than a single source of variation satisfying an absolute exclusion condition: the substantive bar is generally regarded as lower than the instrument's strict exclusion, and the identification rests on the pair as a whole. The cost, as the two applications above illustrate, is a high validation burden (each variable must be classified as W , Z , or X and the bridge plausibility defended variable by variable) and a fragility when the framework is imposed post-hoc on datasets that were not collected with proximal identification in mind. The natural response is to design future studies that anticipate it.

8.6 Outlook

Applying PCI a posteriori to a dataset not designed with proximal identification in mind is operationally challenging: credible W or Z are unlikely unless the dataset already contains them. The two applications above sit at this boundary, and the limits they expose (single-proxy envelopes, post-hoc W/Z classification, partial Assumption 2.4 violations on the father design) are limitations that any post-hoc analysis must accept.

We propose to design experiments that provide W and Z by construction, which makes the framework operational. Clinical studies offer routinely-collected oximetry, blood gases, and pre-intervention biomarkers in unexposed control cohorts as proxies for the latent severity that shapes both treatment selection and outcome. Education research routinely captures parental socio-economic status, prior-period test scores, and

sibling outcomes at enrolment. In macroeconomics, a comparable jurisdiction not subject to a policy provides a synthetic-control-style W , with a lagged exogenous shock as Z . In every case the cost is two design-time auxiliary measurements rather than a full randomisation.

This extends Lipsitch et al. [2]'s negative-control logic, originally used to detect bias rather than to estimate effects, to embedding the negative controls in the design itself [38]. The blind tuning protocol developed in this essay lets us perform honest inference once such a design is available: it calibrates without seeing the truth and validates on disjoint seeds. The lengthscale remains the one parameter the analyst must anchor externally. The methods are in place. What we need next is to design future studies around them.

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